

```
<!DOCTYPE html>
<html lang="zh-TW">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>互動式歷史實驗室 (Interactive History Lab) — 修正版</title>
  <!-- Tailwind CSS -->
  <script src="https://cdn.tailwindcss.com"></script>
  <!-- Leaflet CSS & JS -->
  <link rel="stylesheet" href="https://unpkg.com/leaflet@1.9.4/dist/leaflet.css" />
  <script src="https://unpkg.com/leaflet@1.9.4/dist/leaflet.js"></script>
  <!-- Font Awesome -->
  <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/font-awesome@6.4.0/css/all.min.css">
  <script>
    tailwind.config = {
      theme: {
        extend: {
          colors: {
            dark: '#121212',
            darker: '#0a0a0a',
            panel: '#1e1e1e',
            primary: '#d4af37',
            secondary: '#4b5563'
          }
        }
      }
    }
  </script>
  <style>
    body { font-family: 'Noto Sans TC', sans-serif; background-color: #0a0a0a; color:
#e5e5e5; }
    /* Map Customization */
```

```

#map { background-color: #1a202c; }

.leaflet-popup-content-wrapper { background-color: #1e1e1e; color: #fff; border:
1px solid #d4af37; border-radius: 8px;}

.leaflet-popup-tip { background-color: #1e1e1e; }

/* Scrollbar Customization - 確保捲軸明顯易拉 */

::-webkit-scrollbar { width: 12px; height: 12px; }
::-webkit-scrollbar-track { background: #1a1a1a; border-radius: 6px; }
::-webkit-scrollbar-thumb { background: #4b5563; border-radius: 6px; border: 2px
solid #1a1a1a; }

::-webkit-scrollbar-thumb:hover { background: #d4af37; }

/* Animations */

.fade-in { animation: fadeIn 0.3s ease-in-out; }
@keyframes fadeIn { from { opacity: 0; } to { opacity: 1; } }

/* Chat bubbles */

.chat-bubble-ai { background-color: #2d3748; border-left: 4px solid #d4af37; }
.chat-bubble-user { background-color: #4a5568; border-right: 4px solid #63b3ed; }

/* Modal */

.modal-bg { background-color: rgba(0, 0, 0, 0.7); backdrop-filter: blur(4px); }

/* 時間軸「選中事件」的持續高亮（雙向連動 brushing & linking） */
#timeline-tracks-wrapper .tl-ev-selected .tl-dot {
    background: #ffffff !important;
    box-shadow: 0 0 4px rgba(255,255,255,0.45), 0 0 16px
rgba(212,175,55,1) !important;
}
#timeline-tracks-wrapper .tl-ev-selected .tl-label {
    border-color: #d4af37 !important;
    background: rgba(212,175,55,0.18) !important;
}

```

```

    /* 目前年份「播放頭」— 讓時間軸與地圖狀態時時對齊 */
    #timeline-playhead { transition: left 0.25s ease-out; }
  </style>
</head>
<body class="h-screen flex overflow-hidden flex-col md:flex-row">

  <!-- Sidebar / Navigation -->
  <aside id="main-sidebar" class="w-full md:w-64 bg-panel border-b md:border-b-0
md:border-r border-gray-800 flex flex-row md:flex-col justify-between items-center
md:items-start p-4 transition-all duration-300 z-50 shrink-0">
    <div class="flex items-center justify-between w-full mb-0 md:mb-8">
      <div class="flex items-center gap-3">
        <div class="w-10 h-10 rounded-full bg-primary flex items-center justify-
center text-dark font-bold text-xl shrink-0 shadow-[0_0_15px_rgba(212,175,55,0.4)]">
          <i class="fa-solid fa-hourglass-half"></i>
        </div>
        <h1 id="sidebar-title" class="font-bold text-lg hidden md:block tracking-
wider text-primary whitespace-nowrap">歷史實驗室</h1>
      </div>
      <button onclick="toggleSidebar()" class="hidden md:block text-gray-400
hover:text-white transition px-2">
        <i id="sidebar-icon" class="fa-solid fa-chevron-left"></i>
      </button>
    </div>

    <nav class="flex md:flex-col gap-2 w-full">
      <button onclick="switchTab('map-view')" class="nav-btn active flex items-
center gap-3 w-full p-3 rounded-lg bg-gray-800 text-primary hover:bg-gray-800 transition-
colors" data-target="map-view">
        <i class="fa-solid fa-map-location-dot w-6 text-center"></i>
        <span class="nav-text hidden md:block font-medium whitespace-
nowrap">多軌時空地圖</span>
      </button>

```

```
<button onclick="switchTab('ai-view')" class="nav-btn flex items-center gap-3
w-full p-3 rounded-lg text-gray-400 hover:bg-gray-800 hover:text-white transition-colors"
data-target="ai-view">
```

```
<i class="fa-solid fa-robot w-6 text-center"></i>
```

```
<span class="nav-text hidden md:block font-medium whitespace-
nowrap">AI 探究引擎</span>
```

```
</button>
```

```
<button onclick="switchTab('social-view')" class="nav-btn flex items-center
gap-3 w-full p-3 rounded-lg text-gray-400 hover:bg-gray-800 hover:text-white transition-
colors" data-target="social-view">
```

```
<i class="fa-solid fa-users-rays w-6 text-center"></i>
```

```
<span class="nav-text hidden md:block font-medium whitespace-
nowrap">公共歷史沙龍</span>
```

```
</button>
```

```
</nav>
```

```
<div id="sidebar-version" class="mt-auto hidden md:block w-full">
```

```
<div class="text-xs text-gray-500 text-center lg:text-left whitespace-
nowrap">v10.0 Pro Scale & Safe Map</div>
```

```
</div>
```

```
</aside>
```

```
<!-- Main Content Area -->
```

```
<main class="flex-1 relative bg-darker overflow-hidden">
```

```
<!-- MODULE 1 & 2: Map & Timeline View -->
```

```
<section id="map-view" class="view-section h-full flex flex-col fade-in relative">
```

```
<!-- 上半部 (60%)：地圖區塊 -->
```

```
<div style="flex: 6;" class="flex flex-col relative z-0 shrink-0 border-b border-
gray-800 shadow-xl">
```

```
<!-- Toolbar -->
```

```

<div class="h-14 bg-panel border-b border-gray-800 flex items-center px-
4 justify-between shrink-0 shadow-sm z-10">
  <div class="flex gap-2">
    <button onclick="toggleClimateLayer()" class="px-3 py-1.5
bg-gray-800 hover:bg-gray-700 rounded text-sm text-gray-300 border border-gray-600
transition shadow hidden md:block">
      <i class="fa-solid fa-cloud-sun-rain mr-1"></i> 氣候巨變
圖層
    </button>
    <button onclick="drawTradeRoutes()" class="px-3 py-1.5 bg-
gray-800 hover:bg-gray-700 rounded text-sm text-gray-300 border border-gray-600 transition
shadow">
      <i class="fa-solid fa-route mr-1"></i> 全球貿易網絡
    </button>
  </div>

  <div class="text-sm text-primary font-mono font-bold bg-dark px-4
py-2 rounded-lg border border-gray-700 shadow-inner flex items-center gap-1">
    <span class="hidden md:inline">地圖年份:</span>
    <input type="number" id="current-year-input" class="bg-gray-
900 border border-gray-600 text-primary text-xl font-bold px-2 py-0.5 rounded w-28 text-
center focus:outline-none focus:border-primary transition shadow-inner" value="1945">
    <span id="bc-ad-indicator" class="text-xs px-1.5 py-0.5 bg-
blue-950 text-blue-200 rounded">西元後</span>
  </div>
</div>

<!-- Map Container -->
<div id="map" class="flex-1 w-full relative z-0"></div>
</div>

<!-- 下半部 (40%)：時間軸區塊 -->

```

```

<div style="flex: 4;" class="bg-panel border-t-2 border-primary flex flex-col
relative z-20 shadow-[0_-5px_20px_rgba(0,0,0,0.6)] shrink-0 min-h-[250px]">
  <div class="p-3 border-b border-gray-800 flex flex-col md:flex-row
justify-between items-start md:items-center bg-gray-900 gap-4 shrink-0 shadow-md z-30">
    <div class="flex flex-col gap-1.5">
      <div class="flex items-center gap-3">
        <span class="text-sm font-bold text-gray-200"><i
class="fa-solid fa-timeline mr-2 text-primary"></i>多軌並行時空軸</span>
        <div id="track-filters" class="flex gap-1.5">
          <button onclick="toggleTrack('世界')" data-track="
世界" class="track-btn bg-primary text-dark font-bold text-xs px-2 py-0.5 rounded shadow
transition-colors">世界</button>
          <button onclick="toggleTrack('中國')" data-track="
中國" class="track-btn bg-primary text-dark font-bold text-xs px-2 py-0.5 rounded shadow
transition-colors">中國</button>
          <button onclick="toggleTrack('日本')" data-track="
日本" class="track-btn bg-primary text-dark font-bold text-xs px-2 py-0.5 rounded shadow
transition-colors">日本</button>
          <button onclick="toggleTrack('臺灣')" data-track="
臺灣" class="track-btn bg-primary text-dark font-bold text-xs px-2 py-0.5 rounded shadow
transition-colors">臺灣</button>
        </div>
      </div>
      <p class="text-[11px] text-gray-400"><i class="fa-solid fa-
arrows-up-down-left-right mr-1"></i>可自由拖動檢視；點擊上方標籤可篩選顯示地區
</p>
    </div>
    <div class="flex items-center w-full md:w-1/2 gap-3 bg-darker p-2
rounded-lg border border-gray-700 shadow-inner">
      <span class="text-xs font-bold text-gray-500">前 5000</span>
      <input type="range" id="timeline-slider" min="0" max="1000"
value="0" class="w-full accent-primary cursor-pointer hover:accent-yellow-400 transition">
      <span class="text-xs font-bold text-gray-500">2050</span>
    </div>
  </div>

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        </div>
    </div>

    <div id="timeline-scroll-container" class="flex-1 overflow-auto relative
custom-scrollbar bg-darker">
        <div id="timeline-tracks-wrapper" class="flex flex-col pb-8 pt-2">
            <!-- Dynamic Tracks + Playhead injected by JS -->
        </div>
    </div>

    <!-- 事件詳情面板 -->
    <aside id="event-panel" class="absolute inset-y-0 right-0 w-full sm:w-[380px]
bg-panel border-l-2 border-primary shadow-[-8px_0_30px_rgba(0,0,0,0.6)] z-[1200]
translate-x-full transition-transform duration-300 flex flex-col">
        <div class="p-5 border-b border-gray-800 bg-gray-900 relative shrink-
0">
            <button onclick="closeEventPanel()" class="absolute top-4 right-4
text-gray-400 hover:text-white text-xl transition-transform hover:rotate-90"><i class="fa-
solid fa-xmark"></i></button>

            <span class="text-xs font-bold text-dark bg-primary px-3 py-1
rounded-full mb-3 inline-block shadow">歷史節點</span>

            <h2 id="panel-title" class="text-xl font-bold text-white mb-3 pr-6
leading-snug tracking-wide">事件標題</h2>

            <div class="flex flex-wrap items-center text-xs text-gray-400 gap-
2">
                <span id="panel-time" class="bg-gray-800 px-2 py-1
rounded"><i class="fa-regular fa-clock mr-1 text-primary"></i> 年份</span>

                <span id="panel-location" class="bg-gray-800 px-2 py-1
rounded"><i class="fa-solid fa-location-dot mr-1 text-blue-400"></i> 地點</span>
            </div>
        </div>
    </div>

    <div class="p-5 bg-darker overflow-y-auto flex-1 custom-scrollbar">

```

```
<p id="panel-desc" class="text-gray-300 leading-relaxed mb-6
border-l-4 border-gray-700 pl-4">事件描述...</p>

<div class="bg-gray-900 rounded-lg p-4 border border-gray-700
shadow-inner">

    <h3 class="text-primary font-bold mb-3 flex items-center"><i
class="fa-solid fa-shuffle mr-2"></i>蝴蝶效應 (連鎖影響)</h3>

    <ul id="panel-effects" class="list-disc list-outside ml-5 text-
gray-300 space-y-2 text-sm marker:text-primary"></ul>

</div>

</div>

<div class="p-4 border-t border-gray-800 bg-panel shrink-0 space-y-3">

    <button onclick="discussSelectedEvent()" class="w-full bg-gray-
800 hover:bg-gray-700 text-primary border border-primary/60 px-3 py-2.5 rounded-lg text-
sm font-bold transition shadow"><i class="fa-solid fa-comments mr-2"></i>在歷史沙龍討
論此事件</button>

    <div>

        <div class="text-xs text-gray-400 font-bold mb-2 text-
center"><i class="fa-solid fa-robot mr-1"></i> 帶入 AI 探究模式</div>

        <div class="flex gap-2">

            <button onclick="exploreInAI('socratic')" class="flex-1
bg-gray-700 hover:bg-gray-600 text-white px-2 py-2 rounded-lg text-xs transition shadow
border border-gray-600">蘇格拉底</button>

            <button onclick="exploreInAI('roleplay')" class="flex-1
bg-gray-700 hover:bg-gray-600 text-white px-2 py-2 rounded-lg text-xs transition shadow
border border-gray-600">角色扮演</button>

            <button onclick="exploreInAI('debate')" class="flex-1 bg-
gray-700 hover:bg-gray-600 text-white px-2 py-2 rounded-lg text-xs transition shadow border
border-gray-600">爭議辯論</button>

        </div>

    </div>

</div>

</aside>

</section>
```



```
<!-- MODULE 3: AI Interaction Engine -->

<section id="ai-view" class="view-section h-full hidden flex-col fade-in bg-dark">
  <div class="flex h-full">
    <!-- AI Personas Sidebar -->
    <div class="w-64 border-r border-gray-800 bg-panel p-4 flex flex-col
gap-4 shrink-0 hidden md:flex">
      <h2 class="font-bold text-lg text-primary border-b border-gray-700
pb-2">選擇探究模式</h2>

      <button onclick="setAIPersona('socratic')" class="persona-btn
border-primary bg-gray-700 p-3 rounded-lg text-left hover:border-primary border transition
shadow">

        <div class="font-bold text-gray-200"><i class="fa-solid fa-
masks-theater text-primary mr-2"></i>蘇格拉底教練</div>

        <div class="text-xs text-gray-400 mt-1">引導思考，不直接給
答案，幫助解開盲點。</div>

      </button>

      <button onclick="setAIPersona('roleplay')" class="persona-btn bg-
gray-800 p-3 rounded-lg text-left hover:border-primary border border-transparent transition
shadow">

        <div class="font-bold text-gray-200"><i class="fa-solid fa-
user-tie text-blue-400 mr-2"></i>歷史關鍵人物扮演</div>

        <div class="text-xs text-gray-400 mt-1">動態 Cosplay！根據
您探討的事件，AI 自動切換為當時的各國領袖或官員。</div>

      </button>

      <button onclick="setAIPersona('debate')" class="persona-btn bg-
gray-800 p-3 rounded-lg text-left hover:border-primary border border-transparent transition
shadow">

        <div class="font-bold text-gray-200"><i class="fa-solid fa-
gavel text-red-400 mr-2"></i>歷史爭議辯論庭</div>
```

<div class="text-xs text-gray-400 mt-1">探討如當代文物返還、地緣政治、陣營對立等思辨議題。</div>

</button>

</div>

<!-- Chat Area -->

<div class="flex-1 flex flex-col bg-darker">

<div class="p-4 border-b border-gray-800 bg-panel shadow-sm">

<h3 id="chat-title" class="font-bold text-xl text-gray-200">當前模式：蘇格拉底歷史教練</h3>

</div>

<div id="chat-box" class="flex-1 overflow-y-auto p-6 flex flex-col gap-4">

<div class="flex justify-start fade-in">

<div class="max-w-[80%] chat-bubble-ai p-4 rounded-xl rounded-tl-none text-gray-100 shadow-md flex gap-3">

<i class="fa-solid fa-robot mt-1 text-primary shrink-0"></i>

<div>系統已載入【蘇格拉底歷史教練】設定。請從時空地圖點擊「帶入 AI」，或直接輸入你想探究的古今議題。</div>

</div>

</div>

</div>

<div class="p-4 bg-panel border-t border-gray-800">

<div class="flex gap-2 relative">

<textarea id="ai-input" rows="2" class="w-full bg-gray-900 border border-gray-700 rounded-lg p-3 text-white focus:outline-none focus:border-primary resize-none shadow-inner" placeholder="輸入你的論點或問題 (支援 Shift+Enter 換行)..."></textarea>

```

        <button onclick="sendAIMessage()" class="bg-primary
hover:bg-yellow-600 text-dark font-bold px-6 py-2 rounded-lg transition-colors flex items-
center justify-center shrink-0 shadow-lg">
            傳送 <i class="fa-solid fa-paper-plane ml-2"></i>
        </button>
    </div>
    <div id="ai-loading" class="hidden text-xs text-primary mt-2
animate-pulse">AI 正在深度思考中...</div>
    <div id="ai-key-note" class="hidden text-[11px] text-amber-
400/80 mt-2"><i class="fa-solid fa-circle-info mr-1"></i>提示：此頁面在 Gemini Canvas /
AI Studio 環境會自動注入金鑰；若獨立開啟，需於程式碼的 <code>apiKey</code> 填
入你的 Gemini API Key 才能對話。</div>
</div>
</div>
</div>
</section>

<!-- MODULE 4: Social Forum -->
<section id="social-view" class="view-section h-full hidden fade-in bg-dark p-6
overflow-y-auto">
    <div class="max-w-5xl mx-auto">
        <div class="flex flex-col md:flex-row justify-between md:items-end mb-8
border-b border-gray-700 pb-4 gap-4">
            <div>
                <h2 class="text-3xl font-bold text-primary mb-2">啟蒙歷史沙
龍 <i class="fa-solid fa-mug-hot"></i></h2>
                <p class="text-gray-400">理性思辨的公共領域，針對古今各
項歷史爭議議題進行思辨交流。</p>
            </div>
            <div class="flex items-center gap-3 shrink-0">
                <!-- 身分設定 -->
                <div id="identity-bar" class="flex items-center gap-2 bg-panel
border border-gray-700 rounded-full px-3 py-1.5 shadow"></div>

```

```
        <button onclick="openNewPostModal()" class="bg-primary
hover:bg-yellow-600 text-dark font-bold px-4 py-2 rounded shadow-lg transition whitespace-
nowrap">
```

```
        <i class="fa-solid fa-pen-fancy mr-1"></i> 發起新議題
```

```
    </button>
```

```
</div>
```

```
</div>
```

```
<div id="forum-grid" class="grid grid-cols-1 md:grid-cols-2 gap-6 pb-
10">
```

```
    <!-- 由 JS 依資料 (含 localStorage) 動態產生 -->
```

```
</div>
```

```
</div>
```

```
</section>
```

```
</main>
```

```
<!-- Modals -->
```

```
<div id="forum-write-modal" class="fixed inset-0 modal-bg hidden items-center justify-
center z-[9990] px-4">
```

```
    <div class="bg-panel border border-gray-700 rounded-xl max-w-xl w-full shadow-
2xl relative flex flex-col p-6">
```

```
        <h2 class="text-2xl font-bold text-primary mb-4 border-b border-gray-800 pb-
2"><i class="fa-solid fa-pen-to-square mr-2"></i>發起新議題</h2>
```

```
        <label class="text-sm text-gray-400 mb-1">議題標題</label>
```

```
        <input type="text" id="new-post-title" class="w-full bg-gray-900 border
border-gray-700 rounded-lg p-3 text-white focus:outline-none focus:border-primary mb-4
shadow-inner" placeholder="請輸入討論標題...">
```

```
        <label class="text-sm text-gray-400 mb-1">標籤分類</label>
```

```
        <select id="new-post-tag" class="w-full bg-gray-900 border border-gray-700
rounded-lg p-3 text-white focus:outline-none focus:border-primary mb-4 shadow-inner">
```

```
            <option value="歷史思辨">歷史思辨</option>
```

```
            <option value="文明起源">文明起源</option>
```

```

        <option value="帝國擴張">帝國擴張</option>
        <option value="轉型正義">轉型正義</option>
        <option value="現代局勢">現代局勢</option>
    </select>

    <label class="text-sm text-gray-400 mb-1">您的論述內容</label>
    <textarea id="new-post-content" rows="5" class="w-full bg-gray-900 border
border-gray-700 rounded-lg p-3 text-white focus:outline-none focus:border-primary resize-
none mb-6 shadow-inner" placeholder="請理性闡述您的觀點..."></textarea>

    <div class="flex justify-end gap-4">
        <button onclick="closeModal('forum-write-modal')" class="text-gray-400
hover:text-white px-4 py-2 font-medium">取消</button>

        <button onclick="submitNewPost()" class="bg-primary hover:bg-yellow-
600 text-dark font-bold px-6 py-2 rounded-lg transition shadow-lg">發布議題</button>
    </div>
</div>
</div>
</div>

<div id="forum-read-modal" class="fixed inset-0 modal-bg hidden items-center justify-
center z-[9990] px-4">
    <div class="bg-panel border border-gray-700 rounded-xl max-w-2xl w-full max-h-
[90vh] overflow-y-auto shadow-2xl relative flex flex-col">
        <button onclick="closeModal('forum-read-modal')" class="absolute top-4
right-4 text-gray-400 hover:text-white text-2xl transition-transform hover:rotate-90"><i
class="fa-solid fa-xmark"></i></button>

        <div class="p-6 border-b border-gray-800 bg-gray-900 rounded-t-xl">
            <span id="read-post-tag" class="text-xs px-2 py-1 rounded mb-3 inline-
block">標籤</span>

            <h2 id="read-post-title" class="text-2xl font-bold text-white mb-2
leading-snug">議題標題</h2>

            <div class="flex items-center text-sm text-gray-500 gap-2 mt-3 bg-darker
inline-flex px-3 py-1.5 rounded-lg border border-gray-800">
                <div class="w-6 h-6 rounded-full bg-gray-600 flex items-center
justify-center text-white text-xs shadow-inner" id="read-post-avatar">U</div>

```

```
<span id="read-post-author" class="font-bold text-gray-
300">Author</span>

<span class="mx-1 text-gray-600">|</span><span id="read-post-
time">剛剛</span>

</div>

</div>

<div class="p-6 bg-darker">

  <p id="read-post-content" class="text-gray-300 leading-relaxed text-lg
mb-8 tracking-wide whitespace-pre-wrap">內容...</p>

  <h3 class="text-sm font-bold text-primary mb-4 border-b border-gray-
800 pb-2"><i class="fa-regular fa-comments mr-2"></i>留言思辨區</h3>

  <div class="space-y-4 mb-6" id="mock-comments">

    <!-- 留言由 JS 依該議題的資料動態生成 -->

  </div>

  <div class="flex gap-3 bg-gray-900 p-3 rounded-xl border border-gray-
700 shadow-inner">

    <input type="text" id="forum-reply-input" class="flex-1 bg-
transparent border-none p-2 text-sm text-white focus:outline-none placeholder-gray-500"
placeholder="發表您的回覆 (按 Enter 傳送)...">

    <button onclick="submitForumReply()" class="bg-gray-800 text-
primary border border-gray-600 px-5 py-2 rounded-lg text-sm font-bold hover:bg-gray-700
transition shadow">回覆</button>

  </div>

</div>

</div>

</div>

<!-- 身分設定 Modal -->

<div id="identity-modal" class="fixed inset-0 modal-bg hidden items-center justify-
center z-[9999] px-4">

  <div class="bg-panel border border-gray-700 rounded-xl max-w-md w-full
shadow-2xl relative flex flex-col p-6">
```

```
<h2 class="text-2xl font-bold text-primary mb-2"><i class="fa-solid fa-id-  
badge mr-2"></i>設定你的沙龍身分</h2>
```

```
<p class="text-xs text-gray-400 mb-4 leading-relaxed">這個名字會顯示在你  
發表的議題與留言上。<br>身分目前僅儲存在<span class="text-gray-200 font-bold">這  
台裝置的瀏覽器</span>中（若要跨裝置、多人共用，需接後端如 Firebase）。</p>
```

```
<label class="text-sm text-gray-400 mb-1">顯示名稱</label>
```

```
<input type="text" id="identity-name-input" maxlength="20" class="w-full  
bg-gray-900 border border-gray-700 rounded-lg p-3 text-white focus:outline-none  
focus:border-primary mb-2 shadow-inner" placeholder="例如：思辨中的歷史人 / 清廷觀  
點代言人...">
```

```
<p class="text-[11px] text-gray-500 mb-6">小技巧：你也可以取一個「歷史  
視角」名稱，用某個立場參與討論（歷史觀點取替）。</p>
```

```
<div class="flex justify-end gap-4">
```

```
<button onclick="closeIdentityModal()" class="text-gray-400 hover:text-  
white px-4 py-2 font-medium">取消</button>
```

```
<button onclick="saveIdentityFromModal()" class="bg-primary  
hover:bg-yellow-600 text-dark font-bold px-6 py-2 rounded-lg transition shadow-lg">儲存身  
分</button>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<div id="toast" class="fixed bottom-6 right-6 bg-gray-900 border-l-4 border-primary  
text-white px-6 py-4 rounded shadow-[0_5px_15px_rgba(0,0,0,0.5)] transform translate-y-20  
opacity-0 transition-all duration-300 z-[10000] flex items-center">
```

```
<i class="fa-solid fa-circle-info text-primary mr-3 text-lg"></i>
```

```
<span id="toast-msg">Message</span>
```

```
</div>
```

```
<script>
```

```
// 註：在 Gemini Canvas / Google AI Studio 環境會自動注入金鑰。獨立開啟  
時請填入自己的 Gemini API Key。
```

```
const apiKey = "";
```

```
let currentPersona = null;
let aiChatHistory = [];

// 統一狀態 —— 全域「目前年份」與「選中事件」為單一真相來源
let currentYear = 1945;
let selectedEvent = null;
let selectedEventId = null;

// 狀態：目前啟用的時間軸軌道
let activeTracks = ["世界", "中國", "日本", "臺灣"];

// --- 1. 全球歷史大事記資料庫 ---
const historyData = [
  // 史前與上古
  { id: "human_origin", year: -3900, track: "世界", title: "史前文明起源與採集", lat: 9.0, lng: 38.7, desc: "早期人類依賴舊石器打製石器，已知用火、馴化家犬，並逐步發展出農耕。", effects: ["開啟了對自然資源的最初步馴化。"] },
  { id: "uruk_city", year: -3500, track: "世界", title: "烏魯克城市文明與楔形文字", lat: 31.3, lng: 45.6, desc: "兩河流域進入烏魯克時期，創造世界上最早的文字——楔形文字。", effects: ["文字發明使人類能跨越時空積累知識。"] },
  { id: "hammurabi", year: -1792, track: "世界", title: "漢穆拉比法典頒布", lat: 32.5, lng: 44.4, desc: "古巴比倫國王漢穆拉比制定《漢穆拉比法典》。", effects: ["實行『以眼還眼』的同態復仇原則。"] },
  { id: "socrates", year: -469, track: "世界", title: "蘇格拉底在世", lat: 37.9, lng: 23.7, desc: "首創『蘇格拉底提問法』，將哲學探究核心轉向人類道德與心靈。", effects: ["奠定西方理性批判基礎。"] },
  { id: "roman_empire", year: -27, track: "世界", title: "屋大維與羅馬帝國", lat: 41.9, lng: 12.5, desc: "屋大維被尊為奧古斯都，羅馬進入帝國時期。", effects: ["羅馬法深遠影響近代西方。"] },
  { id: "xia_dynasty", year: -2070, track: "中國", title: "夏朝建立", lat: 34.7, lng: 112.8, desc: "傳說中中國第一個世襲王朝，以黃河流域為核心地區。", effects: ["開啟了『家天下』的世襲政治制度。"] },
```


{ id: "shang_dynasty", year: -1600, track: "中國", title: "商朝建立", lat: 36.1, lng: 114.3, desc: "中國第一個有直接文字記載（甲骨文）的王朝，青銅文明發達。", effects: ["甲骨文奠定漢字基礎，宗教祭祀影響深遠。"] },

{ id: "zhou_dynasty", year: -1046, track: "中國", title: "武王伐紂建周（西周）", lat: 34.2, lng: 108.9, desc: "武王伐紂滅商，建立周朝，實施宗法與封建制度。", effects: ["奠定中國早期封建政治與禮樂文化基礎。"] },

{ id: "spring_autumn", year: -770, track: "中國", title: "平王東遷與春秋時期", lat: 34.7, lng: 112.4, desc: "周平王東遷洛邑，王權衰落，進入春秋五霸諸侯爭霸時期。", effects: ["封建制度逐漸鬆動。"] },

{ id: "warring_states", year: -475, track: "中國", title: "戰國時期", lat: 34.7, lng: 112.4, desc: "三家分晉，進入戰國七雄兼併戰爭的時期，各國紛紛變法圖強。", effects: ["促成了中央集權國家的雛形與百家爭鳴。"] },

{ id: "qin_unite", year: -221, track: "中國", title: "秦滅六國大一統", lat: 34.3, lng: 108.9, desc: "始皇帝開創皇帝制度、推行郡縣制、統一文字與度量衡。", effects: ["確立東亞大陸的中央集權政治架構。"] },

{ id: "western_han", year: -202, track: "中國", title: "劉邦建立西漢", lat: 34.3, lng: 108.9, desc: "劉邦擊敗項羽，建立漢朝，初期實施休養生息與郡國並行制。", effects: ["延續了秦朝的大一統格局並加以鞏固。"] },

{ id: "han_wudi", year: -134, track: "中國", title: "漢武帝盛世與擴張", lat: 34.3, lng: 108.9, desc: "漢武帝罷黜百家獨尊儒術，並對外大舉擴張，將西域、南越、朝鮮半島北部納入版圖。", effects: ["奠定了漢人、漢字、漢文化的深厚基礎，開通絲綢之路。"] },

{ id: "confucius", year: -551, track: "中國", title: "孔子與儒家思想誕生", lat: 35.6, lng: 116.9, desc: "孔子創立儒家學派，首創私人講學。", effects: ["成為兩千年東亞社會控制核心。"] },

{ id: "eastern_han", year: 25, track: "中國", title: "光武中興與東漢建立", lat: 34.7, lng: 112.4, desc: "劉秀重建漢朝，定都洛陽，史稱東漢。", effects: ["豪族勢力逐漸擴大，影響後世社會結構。"] },

{ id: "three_kingdoms", year: 220, track: "中國", title: "三國鼎立", lat: 34.3, lng: 108.9, desc: "曹丕篡漢，隨後劉備、孫權相繼稱帝，天下三分。", effects: ["促進了江南與西南地區的開發。"] },

{ id: "western_jin", year: 280, track: "中國", title: "西晉短暫統一", lat: 34.7, lng: 112.4, desc: "晉滅吳，結束三國分裂，但不久即爆發八王之亂。", effects: ["九品中正制導致世族壟斷政治。"] },

{ id: "yongjia", year: 311, track: "中國", title: "永嘉之禍與衣冠南渡", lat: 34.7, lng: 112.4, desc: "匈奴攻陷洛陽，晉室南遷建立東晉，北方進入五胡十六國。", effects: ["促成中國歷史上第一次大規模人口南遷，加速江南開發。"] },

{ id: "liu_song", year: 420, track: "中國", title: "劉裕建宋（南朝開始）", lat: 32.0, lng: 118.8, desc: "劉裕篡晉建宋，南朝開始，江南經濟持續發展。", effects: ["確立了南朝政權的更迭模式，江南農業與文化繁榮。"] },

{ id: "northern_wei", year: 439, track: "中國", title: "北魏統一北方", lat: 40.1, lng: 113.3, desc: "鮮卑族建立的北魏統一了黃河流域，結束五胡十六國的分裂局面。", effects: ["北方逐漸走向民族融合。"] },

{ id: "xiaowen_reform", year: 493, track: "中國", title: "北魏孝文帝遷都與漢化", lat: 34.7, lng: 112.4, desc: "孝文帝將首都從平城遷至洛陽，並全面推行漢化政策（改漢姓、穿漢服、通漢婚）。", effects: ["極大地促進了胡漢民族的深度融合，為後來隋唐大一統奠定基礎。"] },

{ id: "sui_unite", year: 589, track: "中國", title: "隋朝統一與大運河", lat: 34.3, lng: 108.9, desc: "隋滅陳，結束南北朝近三百年的分裂；開鑿大運河溝通南北。", effects: ["確立三省六部與科舉制，奠定唐朝盛世基礎。"] },

// 中古與近現代

{ id: "jesus_cross", year: 30, track: "世界", title: "耶穌遇難與基督教傳布", lat: 31.7, lng: 35.2, desc: "耶穌被釘十字架，門徒傳布基督教。", effects: ["發展為影響全球的世界性宗教。"] },

{ id: "rome_split", year: 395, track: "世界", title: "羅馬帝國正式分裂", lat: 41.9, lng: 12.5, desc: "帝國正式分裂為東、西羅馬帝國。", effects: ["東羅馬延續千年，西羅馬逐漸瓦解。"] },

{ id: "hegira", year: 622, track: "世界", title: "穆罕默德聖遷", lat: 24.4, lng: 39.6, desc: "逃亡至麥地那，定為伊斯蘭教回曆元年。", effects: ["阿拉伯帝國迅速擴張。"] },

{ id: "crusades_1", year: 1096, track: "世界", title: "第一次十字軍東征", lat: 31.7, lng: 35.2, desc: "教宗號召十字軍東征。", effects: ["促進東西方貿易與文化交流。"] },

{ id: "mongol_empire", year: 1206, track: "世界", title: "成吉思汗建立大蒙古國", lat: 47.9, lng: 106.9, desc: "鐵木真統一蒙古，建立大蒙古國。", effects: ["建立橫跨歐亞的超級帝國。"] },

{ id: "luther_95", year: 1517, track: "世界", title: "馬丁路德與宗教改革", lat: 51.8, lng: 12.6, desc: "張貼九十五條論綱。", effects: ["打破羅馬公教單一權威。"] },

{ id: "magellan", year: 1519, track: "世界", title: "麥哲倫環球航行出發", lat: 36.7, lng: -6.3, desc: "麥哲倫率西班牙船隊尋找香料群島。", effects: ["證明地圓說，開啟全球海洋貿易時代。"] },

{ id: "french_rev", year: 1789, track: "世界", title: "法國大革命爆發", lat: 48.8, lng: 2.3, desc: "巴黎人民攻陷巴士底監獄，推翻專制。", effects: ["自由、平等的思想震撼全歐洲。"] },

{ id: "napoleon", year: 1804, track: "世界", title: "拿破崙稱帝", lat: 48.8, lng: 2.3, desc: "拿破崙加冕為法蘭西第一帝國皇帝。", effects: ["將大革命精神與民法典強制推廣至歐洲各地。"] },

{ id: "tang_dynasty", year: 618, track: "中國", title: "唐朝建立", lat: 34.3, lng: 108.9, desc: "李淵稱帝建立唐朝。", effects: ["發展出極盛的東亞文化圈。"] },

{ id: "tian_kehan", year: 630, track: "中國", title: "平定東突厥與天可汗體系", lat: 41.0, lng: 111.0, desc: "唐太宗擊敗東突厥，被西北各族尊為「天可汗」，疆域廣闊。", effects: ["確立了唐帝國在東亞及中亞的霸主地位，絲路貿易大盛。"] },

{ id: "an_shi", year: 755, track: "中國", title: "安史之亂", lat: 34.3, lng: 108.9, desc: "安祿山、史思明起兵叛唐，唐朝由盛轉衰。", effects: ["導致藩鎮割據，經濟重心進一步南移。"] },

{ id: "khitan_liao", year: 916, track: "中國", title: "契丹建國（遼）", lat: 43.8, lng: 119.2, desc: "耶律阿保機建立契丹國（後改稱遼），並逐步南下擴張。", effects: ["改變了東亞大陸北方的政治板塊。"] },

{ id: "northern_song", year: 960, track: "中國", title: "陳橋兵變與北宋建立", lat: 34.8, lng: 114.3, desc: "趙匡胤黃袍加身，建立宋朝，實行強幹弱枝、重文輕武政策。", effects: ["造就了經濟繁榮與文化藝術的高峰，但軍事較為積弱。"] },

{ id: "chanyuan_treaty", year: 1004, track: "中國", title: "澶淵之盟", lat: 35.7, lng: 115.0, desc: "北宋與遼國簽訂和平條約，以兄弟相稱並餽贈歲幣。", effects: ["維持了宋遼之間長達百餘年的相對和平與貿易往來。"] },

{ id: "jingkang", year: 1127, track: "中國", title: "靖康之禍與南宋偏安", lat: 30.2, lng: 120.1, desc: "金兵攻陷汴京，俘虜徽欽二帝，宋室南渡定都臨安(杭州)。", effects: ["中國經濟重心徹底南移完成。"] },

{ id: "jin_fall", year: 1234, track: "中國", title: "蒙宋聯軍滅金", lat: 33.0, lng: 114.0, desc: "蒙古與南宋聯手攻陷蔡州，金朝滅亡，蒙古隨即南下侵宋。", effects: ["蒙古徹底控制華北，南宋面臨更強大的生存威脅。"] },

{ id: "yuan_founded", year: 1271, track: "中國", title: "忽必烈建立元朝", lat: 39.9, lng: 116.4, desc: "忽必烈取《易經》「大哉乾元」之意，改國號為大元，定都大都。", effects: ["中國歷史上第一個由北方遊牧民族建立的大一統王朝。"] },

{ id: "song_fall", year: 1279, track: "中國", title: "崖山海戰與南宋滅亡", lat: 22.2, lng: 113.1, desc: "元軍在崖山海戰徹底擊敗南宋殘餘水師，陸秀夫背負幼帝投海，南宋滅亡。", effects: ["元朝完成全國大一統，結束了自唐末五代以來數百年的分裂局面。"] },

{ id: "ming_founded", year: 1368, track: "中國", title: "朱元璋建立大明", lat: 32.0, lng: 118.8, desc: "朱元璋在應天府(南京)稱帝，建立明朝，隨後北伐攻陷大都，結束元朝在全國的統治。", effects: ["恢復漢人政權，加強君主專制，廢除丞相制度。"] },

{ id: "yongle_peak", year: 1405, track: "中國", title: "永樂盛世與鄭和下西洋", lat: 39.9, lng: 116.4, desc: "明成祖朱棣派鄭和七下西洋，並將東北(奴兒干都司)、安南(交趾)短期納入版圖，威服四方。", effects: ["確立明朝在東亞的朝貢體系霸主地位，促進海上貿易與文化交流。"] },

{ id: "wokou_raids", year: 1550, track: "中國", title: "嘉靖大倭寇", lat: 29.8, lng: 121.5, desc: "東南沿海因海禁政策引發嚴重走私與武裝衝突，史稱「嘉靖大倭患」，後由戚繼光等人平定。", effects: ["促使明朝有限度開放海禁(隆慶開關)，白銀大量流入中國。"] },

{ id: "ming_fall", year: 1644, track: "中國", title: "明朝滅亡與清兵入關", lat: 39.9, lng: 116.4, desc: "李自成攻入北京，吳三桂引清軍入關。", effects: ["開啟了大清帝國的統治。"] },

{ id: "qing_peak", year: 1760, track: "中國", title: "清朝極盛與十全武功", lat: 43.8, lng: 87.6, desc: "清高宗乾隆帝平定準噶爾與回部，將新疆納入版圖，大清帝國疆域達到秋海棠般的極盛規模。", effects: ["奠定現代中國的領土輪廓，多民族國家進一步鞏固。"] },

{ id: "opium_war", year: 1840, track: "中國", title: "中英鴉片戰爭", lat: 30.2, lng: 122.0, desc: "英國為打開中國市場發動鴉片戰爭。", effects: ["中國門戶洞開，逐漸淪為半殖民地。"] },

{ id: "taiping", year: 1851, track: "中國", title: "太平天國運動", lat: 32.0, lng: 118.8, desc: "洪秀全發動金田起義，建號太平天國，定都天京(南京)，嚴重打擊清朝統治。", effects: ["漢人督撫勢力崛起，地方武裝(湘軍、淮軍)成為清晚期主要軍事力量。"] },

{ id: "sino_jap_war", year: 1894, track: "中國", title: "甲午戰爭", lat: 38.9, lng: 121.6, desc: "清朝在甲午戰爭中敗於日本，簽訂《馬關條約》，割讓臺灣與遼東半島。", effects: ["引發列強瓜分中國狂潮，促使維新變法與革命思想興起。"] },

{ id: "boxer", year: 1900, track: "中國", title: "八國聯軍與義和團", lat: 39.9, lng: 116.4, desc: "義和團打出「扶清滅洋」口號，引發八國聯軍攻陷北京，簽訂《辛丑和約》。", effects: ["清朝徹底淪為半殖民地，巨額賠款重創經濟。"] },

{ id: "xinhai_rev", year: 1911, track: "中國", title: "辛亥革命與中華民國成立", lat: 30.6, lng: 114.3, desc: "武昌起義爆發，隔年推翻清朝統治。", effects: ["結束了中國兩千多年的君主專制，建立亞洲第一個民主共和國。"] },

{ id: "may_fourth", year: 1919, track: "中國", title: "五四運動", lat: 39.9, lng: 116.4, desc: "巴黎和會外交失敗引發北京學生大規模抗議。", effects: ["促進馬克思主義在中國的傳播與新文化運動。"] },

{ id: "northern_exp", year: 1926, track: "中國", title: "國民革命軍北伐", lat: 23.1, lng: 113.2, desc: "蔣中正率領國民革命軍自廣州北伐，擊敗北洋軍閥，名義上統一中國。", effects: ["結束軍閥割據，國民政府定都南京，迎來黃金十年。"] },

{ id: "mukden", year: 1931, track: "中國", title: "九一八事變", lat: 41.8, lng: 123.4, desc: "日本關東軍炸毀南滿鐵路，藉口進佔中國東北，隨後建立滿洲國。", effects: ["中日衝突全面升溫，國際聯盟無力制裁。"] },

{ id: "marco_polo", year: 1937, track: "中國", title: "七七事變與抗日戰爭", lat: 39.8, lng: 116.2, desc: "盧溝橋事變爆發，中國展開全面抗日戰爭，國共第二次合作。", effects: ["中國戰場成為二戰同盟國亞洲主戰場，極大消耗日本國力。"] },

{ id: "prc_founded", year: 1949, track: "中國", title: "中華人民共和國成立", lat: 39.9, lng: 116.4, desc: "中共贏得國共內戰，於北京建政。", effects: ["兩岸分立，冷戰格局在東亞確立。"] },

{ id: "cultural_rev", year: 1966, track: "中國", title: "文化大革命", lat: 39.9, lng: 116.4, desc: "毛澤東發動文化大革命，紅衛兵批鬥走資派。", effects: ["對中國傳統文化與社會經濟造成十年浩劫。"] },

{ id: "reform_opening", year: 1978, track: "中國", title: "改革開放", lat: 22.5, lng: 114.0, desc: "中共十一屆三中全會確立改革開放路線，設立深圳等經濟特區。", effects: ["中國經濟開始高速成長，重新融入全球化體系。"] },

{ id: "tiananmen", year: 1989, track: "中國", title: "六四天安門事件", lat: 39.9, lng: 116.4, desc: "北京爆發大規模民主抗議運動，遭軍隊鎮壓。", effects: ["引發國際制裁，改變中國後續政治發展走向。"] },

{ id: "hk_handover", year: 1997, track: "中國", title: "香港主權移交", lat: 22.3, lng: 114.1, desc: "英國將香港主權移交給中華人民共和國，實施「一國兩制」。", effects: ["標誌著西方帝國主義在中國的殖民歷史正式終結。"] },

{ id: "belt_and_road", year: 2013, track: "中國", title: "一帶一路倡議", lat: 39.9, lng: 116.4, desc: "中國提出「絲綢之路經濟帶」和「21世紀海上絲綢之路」，透過基礎建設投資擴大全球政經影響力。", effects: ["引發國際社會對中國地緣政治擴張的關注與美中戰略競爭。"] },

{ id: "taika_reform", year: 645, track: "日本", title: "大化革新", lat: 34.5, lng: 135.8, desc: "日本推行改革，學習唐朝制度。", effects: ["轉型為中央集權國家。"] },

{ id: "black_ships", year: 1853, track: "日本", title: "黑船來航", lat: 35.2, lng: 139.6, desc: "美國海軍將領培里迫使日本開港。", effects: ["打破鎖國，引發幕末尊王攘夷運動。"] },

{ id: "meiji", year: 1868, track: "日本", title: "明治維新", lat: 35.6, lng: 139.6, desc: "明治天皇發布《五條御誓文》，展開現代化改革。", effects: ["使日本躍升為列強。"] },

{ id: "dutch_tw", year: 1624, track: "臺灣", title: "荷蘭人建熱蘭遮堡", lat: 23.0, lng: 120.2, desc: "建立熱蘭遮堡，作為東亞貿易轉運站。", effects: ["臺灣納入全球資本經濟體系。"] },

{ id: "koxinga", year: 1662, track: "臺灣", title: "鄭氏驅逐荷蘭人", lat: 23.0, lng: 120.2, desc: "鄭成功戰敗荷蘭人，設一府二縣。", effects: ["開啟臺灣漢人社會的典章制度。"] },

{ id: "shimonoseki", year: 1895, track: "臺灣", title: "馬關條約割讓臺灣", lat: 25.0, lng: 121.5, desc: "甲午戰爭清廷戰敗，簽訂《馬關條約》。", effects: ["臺灣進入日本殖民時期。"], specialAction: "show1894War" },

{ id: "feb_28", year: 1947, track: "臺灣", title: "二二八事件", lat: 25.0, lng: 121.5, desc: "查緝私菸引發全臺衝突，遭國民政府派軍鎮壓。", effects: ["造成社會菁英斷層，深刻影響戰後族群關係與政治。"] },

{ id: "taiwan_retreat", year: 1949, track: "臺灣", title: "政府遷臺與全島戒嚴", lat: 25.0, lng: 121.5, desc: "國共內戰局勢逆轉，國民政府退守臺灣並發布戒嚴令。", effects: ["臺灣進入長達 38 年的「白色恐怖」與戒嚴時期。"] },

{ id: "taiwan_un", year: 1971, track: "臺灣", title: "中華民國退出聯合國", lat: 40.7, lng: -74.0, desc: "聯合國大會通過 2758 號決議，中華人民共和國取代中華民國席位。", effects: ["臺灣面臨嚴重外交危機與國際孤立。"] },

{ id: "martial_law_lifted", year: 1987, track: "臺灣", title: "解除戒嚴", lat: 25.0, lng: 121.5, desc: "蔣經國總統宣布解除長達 38 年的戒嚴令，開放黨禁、報禁。", effects: ["臺灣社會迎來自由化與民主化的高速發展。"] },

{ id: "pres_election", year: 1996, track: "臺灣", title: "首次總統直選與臺海危機", lat: 25.0, lng: 121.5, desc: "臺灣舉行首次總統直接民選，中共在臺海發射飛彈恫嚇。", effects: ["標誌臺灣民主化重要里程碑。"] },

// 現代局勢與人權、性別發展

{ id: "nightingale", year: 1854, track: "世界", title: "南丁格爾與近代護理學", lat: 44.9, lng: 34.1, desc: "克里米亞戰爭期間，南丁格爾改善戰地醫院衛生，大幅降低死亡率，成為現代護理學先驅。", effects: ["提升女性在醫療專業的地位，挑戰單一性別職業藩籬。"] },

{ id: "ww1_start", year: 1914, track: "世界", title: "第一次世界大戰爆發", lat: 43.8, lng: 18.4, desc: "塞拉耶佛事件引爆同盟國與協約國的全面戰爭。", effects: ["導致俄、德、奧匈、鄂圖曼四大帝國瓦解，重塑了現代歐洲版圖。"] },

{ id: "sanger_clinic", year: 1916, track: "世界", title: "山額夫人開設節育診所", lat: 40.7, lng: -74.0, desc: "山額夫人在紐約開設首家節育診所，推動生育控制觀念。", effects: ["促進女性身體自主權與現代婦女運動的發展。"] },

{ id: "ww2_start", year: 1939, track: "世界", title: "第二次世界大戰爆發", lat: 52.2, lng: 21.0, desc: "納粹德國入侵波蘭，英法對德宣戰。", effects: ["人類歷史上最慘烈與最具毀滅性的全球戰爭，催生了聯合國與冷戰格局。"] },

{ id: "un_founded", year: 1945, track: "世界", title: "二戰結束與聯合國成立", lat: 40.7, lng: -74.0, desc: "美軍在廣島長崎投下原子彈，日本無條件投降。", effects: ["舊帝國主義崩潰，美蘇兩大強權崛起，冷戰序幕拉開。"] },

{ id: "udhr", year: 1948, track: "世界", title: "世界人權宣言", lat: 48.8, lng: 2.3, desc: "聯合國大會於巴黎通過《世界人權宣言》，確立普世人權標準。", effects: ["展現世界主義思想，成為各國人權保障的重要基準。"] },

{ id: "korean_war", year: 1950, track: "世界", title: "韓戰爆發", lat: 38.0, lng: 127.0, desc: "北韓軍隊越過北緯 38 度線突襲南韓。", effects: ["美國派遣第七艦隊協防臺灣，確立冷戰時期東亞反共島鏈。"] },

{ id: "cuba_missile", year: 1962, track: "世界", title: "古巴飛彈危機", lat: 23.1, lng: -82.3, desc: "蘇聯在古巴部署核飛彈，引發美蘇對峙。", effects: ["人類史上最接近全面核戰爭的時刻。"] },

{ id: "feminism_2nd", year: 1968, track: "世界", title: "第二波婦女運動與性別史", lat: 38.9, lng: -77.0, desc: "受婦女運動影響，重新檢視歷史上性別角色、分工與權力關係。", effects: ["打破性別僵化身分，推動當代性別平權與多元性別發展。"] },

{ id: "apollo_11", year: 1969, track: "世界", title: "人類首次登陸月球", lat: 28.5, lng: -80.5, desc: "美國阿波羅 11 號阿姆斯壯等人成功登月。", effects: ["美蘇太空競賽的高潮，展現了人類科技的極限突破。"] },

{ id: "ussr_collapse", year: 1991, track: "世界", title: "蘇聯解體與冷戰結束", lat: 55.7, lng: 37.6, desc: "戈巴契夫辭職，蘇聯正式解體為 15 個國家。", effects: ["長達半世紀的冷戰結束，美國成為全球唯一超級大國。"] },

{ id: "nine_eleven", year: 2001, track: "世界", title: "美國九一一事件", lat: 40.7, lng: -74.0, desc: "蓋達組織劫持客機摧毀紐約世貿雙子星大樓。", effects: ["引發美國領導的全球反恐戰爭(阿富汗、伊拉克)，深刻改變 21 世紀國際局勢。"] },

{ id: "covid_19", year: 2020, track: "世界", title: "COVID-19 全球疫情", lat: 30.5, lng: 114.3, desc: "新冠病毒引發全球大流行，多國實施封城。", effects: ["全球供應鏈重組，改變人類生活與工作型態，加速數位轉型。"] },

{ id: "ukraine_war", year: 2022, track: "世界", title: "俄烏戰爭爆發", lat: 50.4, lng: 30.5, desc: "俄羅斯對烏克蘭展開全面特別軍事行動。", effects: ["二戰後歐洲最大規模的軍事衝突，引發全球能源與糧食危機。"] },

{ id: "gaza_war", year: 2023, track: "世界", title: "以哈戰爭爆發", lat: 31.5, lng: 34.4, desc: "哈瑪斯突襲以色列，引發加薩走廊嚴重軍事衝突。", effects: ["造成嚴重人道危機，中東地緣政治再度陷入高度緊張。"] },


```
    { id: "iran_israel", year: 2025, track: "世界", title: "中東局勢再升溫 (以伊衝突)", lat: 35.7, lng: 51.4, desc: "中東地緣政治惡化，爆發以色列與伊朗的軍事衝突。", effects: ["衝擊全球石油市場與大國博弈。"] }  
  ];
```

```
// --- 2. 帝國疆域座標 ---
```

```
    const romanRepublicCoords = [[[45,8], [46,13], [40,18], [38,16], [37,12], [42,10]],  
    [[41,19], [42,24], [36,22], [38,20]], [[37,9], [37,11], [33,12], [34,8]], [[38,12], [38,15],  
    [36,15], [36,12]]];
```

```
    const romanEmpireCoords = [[[45,8], [46,13], [40,18], [38,16], [37,12], [42,10]],  
    [[41,19], [42,24], [36,22], [38,20]], [[37,9], [37,11], [33,12], [34,8]], [[38,12], [38,15],  
    [36,15], [36,12]], [[43,-9], [44,2], [36,-5], [37,-9]], [[51,-3], [51,7], [44,7], [43,-1]], [[55,-4],  
    [55,1], [51,1], [51,-4]], [[41,26], [41,40], [33,36], [31,34], [36,28]], [[31,29], [31,34], [24,34],  
    [24,29]], [[33,12], [31,29], [28,20], [30,10]], [[36,-5], [36,8], [33,8], [34,-5]]];
```

```
    const romanItaly = [[[45,8], [46,13], [40,18], [38,16], [37,12], [42,10]]];  
    const romanPunic = [...romanItaly, [[37,9], [37,11], [33,12], [34,8]], [[38,12],  
    [38,15], [36,15], [36,12]]];
```

```
    const romanLateRep = romanRepublicCoords;  
    const romanEarlyEmp = romanEmpireCoords;  
    const romanPeakEmp = romanEmpireCoords;
```

```
    const westernRomanCoords = [[[45,8], [46,13], [40,18], [38,16], [37,12], [42,10]],  
    [[43,-9], [44,2], [36,-5], [37,-9]], [[51,-3], [51,7], [44,7], [43,-1]], [[55,-4], [55,1], [51,1], [51,-  
    4]], [[37,9], [37,11], [33,12], [34,8]], [[36,-5], [36,8], [33,8], [34,-5]]];
```

```
    const easternRomanCoords = [[[41,19], [42,24], [36,22], [38,20]], [[41,26], [41,40],  
    [33,36], [31,34], [36,28]], [[31,29], [31,34], [24,34], [24,29]], [[33,12], [31,29], [28,20],  
    [30,10]]];
```

```
    const justinianEmpireCoords = [...easternRomanCoords, [[45,8], [46,13], [40,18],  
    [38,16], [37,12], [42,10]], [[37,9], [37,11], [33,12], [34,8]], [[36,-5], [38,-5], [38,-2], [36,-2]]];
```

```
    const byzantineShrunkCoords = [[[41,19], [42,24], [36,22], [38,20]], [[41,26],  
    [41,40], [36,36], [36,28]], [[38,15], [40,18], [38,16]]];
```

```
const byzantineLateCoords = [[[41.5, 28.5], [41.5, 29.5], [40.5, 29.5], [40.5, 28.5]],  
[[38, 21], [38, 23], [36, 23], [36, 21]]];
```

```
const arabEmpireCoords = [[[30,35], [20,40], [15,45], [25,55], [35,45]], [[35,-5],  
[35,10], [30,35], [20,15], [20,-5]], [[43,-9], [44,2], [36,-5], [37,-9]]];
```

```
const mongolEmpireCoords = [[[50,30], [55,50], [50,80], [55,120], [45,140],  
[30,120], [25,100], [35,70], [30,50], [40,40]]];
```

```
const spanishEmpireCoords = [[[43,-9], [44,2], [36,-5], [37,-9]], [[30,-115], [30,-  
90], [15,-90], [10,-80], [10,-85], [20,-105]], [[10,-75], [-40,-75], [-40,-60], [0,-60]], [[18,120],  
[18,125], [6,125], [6,120]]];
```

```
const britishEmpireCoords = [[[58,-6], [58,2], [50,2], [50,-10]], [[30,70], [30,90],  
[10,80], [10,75]], [[60,-120], [60,-60], [45,-60], [45,-120]], [[-10,115], [-10,150], [-40,150], [-  
40,115]], [[-20, 20], [-35, 20], [-35, 30], [-20, 30]]];
```

```
const egyptNewKingdomCoords = [[[22, 28], [31, 29], [33, 35], [29, 36], [22, 33]]];
```

```
const alexanderEmpireCoords = [[[41, 20], [30, 25], [31, 35], [33, 44], [30, 48], [25,  
60], [30, 75], [40, 70], [42, 60], [42, 30]]];
```

```
// 中國歷代疆域變化 (夏、商、周、秦、漢)
```

```
const xiaDynastyCoords = [[[33, 109], [36, 109], [37, 116], [33, 116]]];
```

```
const shangDynastyCoords = [[[31, 107], [39, 107], [40, 119], [31, 119]]];
```

```
const westernZhouCoords = [[[30, 104], [40, 104], [41, 121], [30, 121]]];
```

```
const easternZhouCoords = [[[29, 103], [42, 103], [42, 122], [29, 122]]];
```

```
const qinDynastyCoords_ancient = [[[21, 104], [25, 121], [42, 123], [42, 104], [35,  
102]]];
```

```
const hanEarlyCoords = [[[24, 104], [28, 121], [42, 123], [42, 104], [35, 102]]];
```

```
const hanWuCoords = [[[15, 105], [22, 120], [40, 128], [42, 110], [40, 95], [42, 75],  
[36, 75], [35, 95]]];
```

```
const easternHanCoords = [[[15, 100], [22, 120], [40, 125], [42, 105], [40, 95], [35,  
95]]];
```

```
const threeKingdomsWei = [[[33, 105], [33, 120], [40, 125], [42, 105]]];
```

```
const threeKingdomsShu = [[[22, 100], [22, 108], [33, 108], [33, 100]]];
```

```

const threeKingdomsWu = [[[20, 108], [22, 120], [33, 120], [33, 108]]];
const westernJinCoords = [[[18, 100], [22, 120], [40, 125], [42, 105], [35, 95]]];

// 南北朝與五胡十六國細化
const easternJinCoords = [[[18, 100], [22, 120], [33, 120], [33, 100]]];
const sixteenKingdomsCoords = [[[33, 100], [33, 120], [42, 125], [42, 95], [38,
95]]];

const southernDynasties = [[[18, 100], [22, 120], [34, 120], [34, 105], [30, 100]]];
const northernDynasties = [[[34, 100], [34, 120], [42, 125], [45, 105]]];

const suiDynastyCoords = [[[18, 100], [22, 120], [40, 125], [42, 105], [35, 95]]];

// 唐朝各時期與周邊政權（吐蕃、回鶻）
const tangEarlyCoords = [[[18, 100], [22, 120], [40, 125], [42, 105], [35, 95]]];
const tangPeakCoords = [[[15, 105], [22, 128], [40, 128], [48, 130], [52, 110], [48,
90], [42, 65], [36, 70], [35, 95]]];
const tangLateCoords = [[[18, 105], [22, 120], [40, 122], [40, 110], [35, 105], [30,
100]]];
const tuboEmpireCoords = [[[27, 85], [27, 100], [32, 105], [38, 105], [40, 95], [35,
80]]];
const tuboMaxCoords = [[[27, 85], [27, 100], [30, 105], [38, 110], [42, 95], [35,
80]]];
const uyghurCoords = [[[42, 95], [40, 110], [45, 125], [52, 115], [50, 90]]];

const fiveDynastiesCoords = [[[20, 100], [22, 120], [39, 118], [39, 105], [35, 100]]];

const northernSongCoords = [[[20, 105], [22, 120], [39, 118], [39, 105], [35, 100]]];
const liaoDynastyCoords = [[[39, 105], [39, 118], [45, 130], [50, 110], [45, 100]]];
const westernXiaCoords = [[[35, 100], [39, 110], [42, 105], [42, 95], [38, 95]]];
const southernSongCoords = [[[18, 105], [22, 120], [33, 120], [33, 105]]];
const jurchenJinCoords = [[[33, 105], [33, 120], [45, 130], [50, 110], [45, 100]]];
const yuanDynastyCoords = [[[18, 105], [22, 120], [40, 130], [55, 135], [55, 100],
[50, 85], [40, 80], [30, 85], [25, 95], [20, 100]]];

```

```

// 大明（常規與永樂擴張）、大清（入關與極盛）

const mingRegularCoords = [[[20, 100], [22, 120], [33, 122], [40, 120], [43, 125],
[43, 115], [40, 105], [35, 95], [25, 98]]];

const mingYongleCoords = [[[15, 108], [20, 100], [22, 120], [33, 122], [40, 120],
[42, 125], [52, 140], [55, 130], [45, 110], [40, 95], [35, 95], [25, 98]]];

const qingEarlyCoords = [[[18, 109], [22, 120], [25, 122], [40, 125], [43, 131], [53,
141], [55, 130], [52, 110], [40, 100], [35, 95], [28, 100], [21, 105]]];

const qingPeakCoords = [[[18, 109], [22, 120], [25, 122], [40, 125], [43, 131], [53,
141], [55, 130], [51, 100], [51, 92], [46, 75], [38, 73], [30, 80], [28, 85], [21, 100]]];

// 晚清(割讓外滿洲)、中華民國、滿洲國、抗日淪陷區、中華人民共和國

const qingLateCoords = [[[18, 109], [22, 120], [25, 122], [40, 125], [43, 131], [48,
135], [50, 120], [51, 100], [51, 92], [46, 75], [38, 73], [30, 80], [28, 85], [21, 100]]];

const rocCoords = qingLateCoords;

const prcCoords = [[[18.2, 109.5], [21.5, 108], [22.5, 114], [25.0, 119], [30.0, 122],
[35.0, 119.5], [37.5, 122.5], [40.0, 124], [42.0, 128], [47.0, 134], [53.5, 123], [49.0, 117],
[45.0, 111], [42.0, 95], [48.0, 87], [40.0, 74], [35.0, 78], [28.0, 86], [29.0, 95], [21.0, 100]]];

const manchukuoCoords = [[[39, 120], [40, 134], [48, 135], [50, 120], [48, 115],
[42, 115]]];

const japaneseOccupiedChina = [[[22, 110], [22, 120], [35, 122], [40, 118], [35,
110], [28, 110]]];

const axisCoords = [
    [[47,5], [55,5], [55,20], [47,20], [40,10]],
    [[30,130], [45,130], [45,145], [30,145], [10,120], [10,100], [20,100]]
];

const ussrCoords = [
    [[50, 20], [70, 20], [75, 100], [65, 180], [45, 135], [40, 50]]
];

```

```
//  
=====
```

```
=  
  
    // 基本 UI 與側邊欄收折  
  
    //  
=====
```

```
=  
  
    let isSidebarCollapsed = false;  
  
    function toggleSidebar() {  
        isSidebarCollapsed = !isSidebarCollapsed;  
        const sidebar = document.getElementById('main-sidebar');  
        const title = document.getElementById('sidebar-title');  
        const texts = document.querySelectorAll('.nav-text');  
        const icon = document.getElementById('sidebar-icon');  
        const version = document.getElementById('sidebar-version');  
  
        if (isSidebarCollapsed) {  
            sidebar.classList.remove('md:w-64');  
            sidebar.classList.add('md:w-20');  
            title.classList.remove('md:block');  
            title.classList.add('md:hidden');  
            texts.forEach(t => { t.classList.remove('md:block');  
t.classList.add('md:hidden'); });  
            icon.classList.remove('fa-chevron-left');  
            icon.classList.add('fa-chevron-right');  
            version.classList.remove('md:block');  
            version.classList.add('md:hidden');  
        } else {  
            sidebar.classList.remove('md:w-20');  
            sidebar.classList.add('md:w-64');  
            title.classList.remove('md:hidden');  
            title.classList.add('md:block');
```

```

        texts.forEach(t => { t.classList.remove('md:hidden');
t.classList.add('md:block'); });

        icon.classList.remove('fa-chevron-right');
        icon.classList.add('fa-chevron-left');
        version.classList.remove('md:hidden');
        version.classList.add('md:block');
    }
    // 等待動畫結束後重新計算地圖大小
    setTimeout(() => { if (map) map.invalidateSize(); }, 300);
}

```

```

function switchTab(tabId) {
    document.querySelectorAll('.view-section').forEach(el =>
el.classList.add('hidden'));
    document.getElementById(tabId).classList.remove('hidden');
    document.querySelectorAll('.nav-btn').forEach(el => {
        el.classList.remove('active', 'bg-gray-800', 'text-primary');
        el.classList.add('text-gray-400');
    });
    const activeBtn = document.querySelector(`.nav-btn[data-target="${tabId}"]`);
    if(activeBtn) {
        activeBtn.classList.remove('text-gray-400');
        activeBtn.classList.add('active', 'bg-gray-800', 'text-primary');
    }
    if (tabId === 'map-view' && map) {
        setTimeout(() => map.invalidateSize(), 100);
    }
}

```

```

function showToast(msg) {
    const toast = document.getElementById('toast');
    document.getElementById('toast-msg').innerText = msg;
    toast.style.opacity = '1';
}

```

```

toast.style.transform = 'translateY(0)';
clearTimeout(showToast._t);
showToast._t = setTimeout(() => {
    toast.style.opacity = '0';
    toast.style.transform = 'translateY(20px)';
}, 4000);
}

function closeModal(modalId) {
    document.getElementById(modalId).classList.add('hidden');
    document.getElementById(modalId).classList.remove('flex');
}

function escapeHtml(s) {
    return String(s).replace(/[\&\<\>"]'/g, c =>
('{'&':'&amp;','<':'&lt;','>':'&gt;','"'&':'&quot;','"'&':'&#39;'}'[c]));
}
// localStorage 安全包裝
const store = {
    get(k){ try { return localStorage.getItem(k); } catch(e){ return null; } },
    set(k,v){ try { localStorage.setItem(k,v); } catch(e){ /* 沙箱封鎖時忽略
*/ } }
};

//
=====
=
// 地圖與時間軸
//
=====
=
let map;
let markers = [];

```

```

let climateLayer = null;
let tradeRoutesLayer = null;
let taiwanLayer = null;
let warAnimationLayer = null;
let empireLayer = null;
let mapPanTimeout = null;

const TIMELINE_MIN_YEAR = -5000;
const TIMELINE_MAX_YEAR = 2050;
// ✨ 大幅增加時間軸總寬度，給予近代史更多伸展空間
const TIMELINE_WIDTH = 12000;
const TRACK_LABEL_WIDTH = 80;

// ✨ 重新設計非線性演算法：近現代(1800 後)將獲得 65% 的超大比例空間
function yearToPercent(year) {
  if (year <= 1000) {
    return ((year - (-5000)) / 6000) * 15; // 前 6000 年佔 15%
  } else if (year <= 1800) {
    return 15 + ((year - 1000) / 800) * 20; // 800 年佔 20%
  } else if (year <= 1900) {
    return 35 + ((year - 1800) / 100) * 25; // 100 年佔 25% (近代)
  } else {
    return 60 + ((year - 1900) / 150) * 40; // 150 年佔 40% (現代，超級展
    開)
  }
}

function percentToYear(percent) {
  if (percent <= 15) {
    return -5000 + (percent / 15) * 6000;
  } else if (percent <= 35) {
    return 1000 + ((percent - 15) / 20) * 800;
  } else if (percent <= 60) {
    return 1800 + ((percent - 35) / 25) * 100;
  }
}

```



```

    } else {
        return 1900 + ((percent - 60) / 40) * 150;
    }
}

// ✨ 新增安全保護：防止 NaN 導致的崩潰
function safeFlyTo(latlng, zoom) {
    if (!map || !map.getSize() || map.getSize().x === 0 || map.getSize().y === 0)
return;

    if (!Array.isArray(latlng) || !Number.isFinite(latlng[0])
|| !Number.isFinite(latlng[1])) return;

    try {
        map.flyTo(latlng, zoom, { animate: true, duration: 1.2 });
    } catch (e) {
        console.warn('Map flyTo bypassed due to sizing issues.');

```

```

    renderTimeline();
    attachTimelineControls();
    goToYear(1945, null, { fly: false });
}

function goToYear(year, focusEvent = null, opts = {}) {
    const { fly = true, debounce = false } = opts;
    if (!Number.isFinite(year)) year = currentYear;
    year = Math.round(year);
    if (year < TIMELINE_MIN_YEAR) year = TIMELINE_MIN_YEAR;
    if (year > TIMELINE_MAX_YEAR) year = TIMELINE_MAX_YEAR;
    currentYear = year;

    const yearInput = document.getElementById('current-year-input');
    const slider = document.getElementById('timeline-slider');
    if (document.activeElement !== yearInput) yearInput.value = year;
    slider.value = yearToPercent(year) * 10;

    const indicator = document.getElementById('bc-ad-indicator');
    if (year < 0) {
        indicator.innerText = "西元前";
        indicator.className = "text-xs px-1.5 py-0.5 bg-red-950 text-red-200
rounded font-bold shadow-inner";
    } else {
        indicator.innerText = "西元後";
        indicator.className = "text-xs px-1.5 py-0.5 bg-blue-950 text-blue-200
rounded font-bold shadow-inner";
    }

    updateMapMarkers(year);
    updateTimelinePlayhead(year);
    const behavior = (fly && !debounce) ? 'smooth' : 'auto';
    scrollTimelineToYear(year, behavior);

```

```

        clearTimeout(mapPanTimeout);
        if (!fly) return;
        if (focusEvent && Number.isFinite(focusEvent.lat) &&
Number.isFinite(focusEvent.lng)) {
            safeFlyTo([focusEvent.lat, focusEvent.lng], 5);
        } else if (debounce) {
            mapPanTimeout = setTimeout(() => flyToYearContext(year), 250);
        } else {
            flyToYearContext(year);
        }
    }
}

function flyToYearContext(year) {
    const valid = historyData.filter(ev => ev && Number.isFinite(ev.lat) &&
Number.isFinite(ev.lng) && activeTracks.includes(ev.track));
    if (!valid.length) return;
    let nearest = valid[0];
    for (const ev of valid) {
        if (Math.abs(ev.year - year) < Math.abs(nearest.year - year)) nearest = ev;
    }
    safeFlyTo([nearest.lat, nearest.lng], 4);
}

function scrollTimelineToYear(year, behavior = 'smooth') {
    const sc = document.getElementById('timeline-scroll-container');
    if (!sc) return;
    const targetX = TRACK_LABEL_WIDTH + (yearToPercent(year) / 100) *
TIMELINE_WIDTH;
    sc.scrollTo({ left: targetX - (sc.clientWidth / 2), behavior });
}

function updateTimelinePlayhead(year) {

```

```

const ph = document.getElementById('timeline-playhead');
if (!ph) return;
ph.style.left = (TRACK_LABEL_WIDTH + (yearToPercent(year) / 100) *
TIMELINE_WIDTH) + 'px';
const flag = document.getElementById('timeline-playhead-flag');
// ✨ 修正：確保文字較長時不會被截斷
if (flag) flag.innerText = year < 0 ? `西元前 ${Math.abs(year)} 年` : `西元
${year} 年`;
}

```

```

function attachTimelineControls() {
const slider = document.getElementById('timeline-slider');
const yearInput = document.getElementById('current-year-input');
slider.addEventListener('input', (e) =>
goToYear(percentToYear(parseFloat(e.target.value) / 10), null, { fly: false }));
slider.addEventListener('change', () => flyToYearContext(currentYear));
yearInput.addEventListener('change', (e) =>
goToYear(parseInt(e.target.value), null, { fly: true }));
yearInput.addEventListener('keydown', (e) => { if (e.key === 'Enter')
{ goToYear(parseInt(e.target.value), null, { fly: true }); e.target.blur(); } });
}

```

```

function toggleTrack(trackName) {
const idx = activeTracks.indexOf(trackName);
if (idx > -1) {
if (activeTracks.length === 1) {
showToast("至少需要保留一個地區時間軸!");
return;
}
activeTracks.splice(idx, 1);
} else {
activeTracks.push(trackName);
}
}

```

```
const order = ["世界", "中國", "日本", "臺灣"];
activeTracks.sort((a, b) => order.indexOf(a) - order.indexOf(b));
```

```
updateTrackFilterUI();
renderTimeline();
updateMapMarkers(currentYear);
updateTimelinePlayhead(currentYear);
}
```

```
function updateTrackFilterUI() {
  document.querySelectorAll('.track-btn').forEach(btn => {
    if (activeTracks.includes(btn.dataset.track)) {
      btn.classList.remove('bg-gray-700', 'text-gray-300');
      btn.classList.add('bg-primary', 'text-dark');
    } else {
      btn.classList.remove('bg-primary', 'text-dark');
      btn.classList.add('bg-gray-700', 'text-gray-300');
    }
  });
}
```

```
function renderTimeline() {
  const container = document.getElementById('timeline-tracks-wrapper');
  container.style.width = `${TIMELINE_WIDTH +
TRACK_LABEL_WIDTH}px`;
  container.style.position = 'relative';

  const tracks = activeTracks;
  let html = "";

  tracks.forEach(track => {
    html += `
```

```

<div class="relative border-b border-dashed border-gray-800 flex"
style="min-height: 180px;">
  <div class="w-20 flex-shrink-0 bg-gray-900 flex items-center
justify-center border-r border-gray-700 shadow-[8px_0_15px_-3px_rgba(0,0,0,0.7)]"
style="position: sticky; left: 0; z-index: 90;">
    <span class="text-primary font-bold text-xl tracking-widest
drop-shadow-md" style="writing-mode: vertical-rl; text-orientation: upright; letter-spacing:
4px;">${track}</span>
  </div>
  <div class="flex-1 relative" style="min-width:
${TIMELINE_WIDTH}px;">
    <div class="absolute top-1/2 left-0 w-full h-px bg-gray-700 -
translate-y-1/2"></div>
  `;

```

```

let trackEvents = historyData.filter(d => d.track === track);
trackEvents.sort((a, b) => a.year - b.year); // 確保照年份排序，以利規
律錯位

```

```

trackEvents.forEach((event, index) => {
  const percent = yearToPercent(event.year);

  // ✨ 修正：4 階層垂直錯位，確保緊密的標籤也不會重疊
  const posIndex = index % 4;
  let topPos, labelStyle, lineStyle;
  if (posIndex === 0) { topPos = '25%'; labelStyle = 'bottom: 16px;';
lineStyle = 'bottom: 0; height: 16px;'; }
  else if (posIndex === 1) { topPos = '40%'; labelStyle = 'bottom:
16px;'; lineStyle = 'bottom: 0; height: 16px;'; }
  else if (posIndex === 2) { topPos = '60%'; labelStyle = 'top: 16px;';
lineStyle = 'top: 0; height: 16px;'; }
  else { topPos = '75%'; labelStyle = 'top: 16px;'; lineStyle = 'top: 0;
height: 16px;'; }
}

```

```

const displayYearText = event.year < 0 ? `前
${Math.abs(event.year)}` : `西元 ${event.year}`;

html += `
  <div id="tl-ev-${event.id}" class="absolute group" style="left:
${percent}%; top: ${topPos};">
    <div class="tl-dot w-3.5 h-3.5 rounded-full bg-primary
border-2 border-darker z-30 absolute cursor-pointer shadow-[0_0_8px_rgba(212,175,55,0.8)]
group-hover:scale-150 group-hover:bg-white transition-all duration-300" style="transform:
translate(-50%, -50%);" onclick="selectEvent('${event.id}')"></div>

    <div class="absolute flex flex-col items-center z-40
transition-all duration-300" style="transform: translateX(-50%); ${labelStyle}">
      <div class="tl-label bg-gray-900/95 border border-
gray-700 text-gray-200 px-2.5 py-1.5 rounded-lg text-center shadow-xl cursor-pointer group-
hover:border-primary ${posIndex < 2 ? 'group-hover:-translate-y-1' : 'group-hover:translate-
y-1'} transition-transform duration-300 w-max max-w-[140px]"
onclick="selectEvent('${event.id}')">

        <div class="text-primary font-bold text-[12px]
mb-0.5">${displayYearText}</div>

        <div class="text-[11px] tracking-wide text-gray-
300 font-medium whitespace-normal leading-snug">${event.title}</div>

      </div>
    </div>
    <div class="absolute w-px bg-gray-600 group-hover:bg-
primary transition-colors duration-300" style="transform: translateX(-50%);
${lineStyle}"></div>
  </div>
`;
});
html += `</div></div>`;
});

```

```

// ✨ 修正：確保播放頭的年份文字不會被截斷
html += `
    <div id="timeline-playhead" style="position:absolute; top:0; bottom:0;
width:2px; background:#d4af37; z-index:60; pointer-events:none; box-shadow:0 0 10px
rgba(212,175,55,0.9); left:${TRACK_LABEL_WIDTH}px;">
        <div id="timeline-playhead-flag" style="position:sticky; top:4px;
left:50%; transform:translateX(-50%); width:max-content; white-space:nowrap;
display:inline-block;" class="bg-primary text-dark text-[11px] font-bold px-2 py-1 rounded
shadow-lg">1945</div>
    </div>
`;

container.innerHTML = html;
}

function selectEvent(eventId) {
    const event = historyData.find(e => e.id === eventId);
    if (!event) return;
    selectedEvent = event;
    selectedEventId = eventId;
    goToYear(event.year, event, { fly: true });
    highlightSelectedTimelineDot(eventId);
    if (event.specialAction) handleSpecialAction(event);
    openEventPanel(event);
}

function highlightSelectedTimelineDot(eventId) {
    document.querySelectorAll('#timeline-tracks-wrapper .tl-ev-
selected').forEach(el => el.classList.remove('tl-ev-selected'));
    const el = document.getElementById('tl-ev-' + eventId);
    if (el) el.classList.add('tl-ev-selected');
}

```



```

function openEventPanel(event) {
    const displayYearText = event.year < 0 ? `西元前 ${Math.abs(event.year)}` :
`西元 ${event.year}`;

    document.getElementById('panel-title').innerText = event.title;
    document.getElementById('panel-time').innerHTML = `

```

```

        if (warAnimationLayer) { map.removeLayer(warAnimationLayer);
warAnimationLayer = null; }

        warAnimationLayer = L.layerGroup().addTo(map);

        const taiwan =
[[25.3,121.5],[24.5,122.0],[22.0,120.9],[22.0,120.2],[23.5,120.0],[24.5,120.5]];

        const liaodong =
[[40.9,121.2],[40.0,122.4],[38.7,121.2],[39.6,121.5],[40.2,120.9]];

        [{c: taiwan, n: '臺灣、澎湖群島'}, {c: liaodong, n: '遼東半島'}].forEach(item
=> {

            L.polygon(item.c, { color: '#ef4444', weight: 3, fillColor: '#ef4444',
fillOpacity: 0.5, dashArray: '6,6', className: 'animate-pulse' })

                .addTo(warAnimationLayer)

                .bindTooltip(`《馬關條約》割讓：${item.n}`, { sticky: true });

        });

        showToast('🇨🇹 1895 《馬關條約》：清廷割讓臺灣、澎湖與遼東半島予日
本');

        setTimeout(() => { if (warAnimationLayer)
{ map.removeLayer(warAnimationLayer); warAnimationLayer = null; } }, 6000);

    }

function updateMapMarkers(year) {
    markers.forEach(m => map.removeLayer(m));
    markers = [];
    year = parseInt(year);

    if (taiwanLayer) {
        if (year >= 1624 && year < 1662) {
            taiwanLayer.setStyle({color: '#f97316', fillColor: '#f97316'});
            taiwanLayer.bindTooltip("<b>臺灣（荷蘭與西班牙統治時
期)</b>");

        } else if (year >= 1662 && year < 1683) {
            taiwanLayer.setStyle({color: '#b91c1c', fillColor: '#b91c1c'});
            taiwanLayer.bindTooltip("<b>臺灣（明鄭時期)</b>");
        }
    }
}

```

```

    } else if (year >= 1683 && year < 1895) {
        taiwanLayer.setStyle({color: '#f59e0b', fillColor: '#f59e0b'});
        taiwanLayer.bindTooltip("<b>臺灣 (清治時期)</b>");
    } else if (year >= 1895 && year < 1945) {
        taiwanLayer.setStyle({color: '#ef4444', fillColor: '#ef4444'});
        taiwanLayer.bindTooltip("<b>臺灣 (日治時期)</b>");
    } else if (year >= 1945) {
        taiwanLayer.setStyle({color: '#3b82f6', fillColor: '#3b82f6'});
        taiwanLayer.bindTooltip("<b>臺灣 (中華民國時期)</b>");
    } else {
        taiwanLayer.setStyle({color: '#22c55e', fillColor: '#22c55e'});
        taiwanLayer.bindTooltip("<b>臺灣 (南島語族與原住民時期)</b>");
    }
}

if (empireLayer) { map.removeLayer(empireLayer); empireLayer = null; }
empireLayer = L.layerGroup().addTo(map);

if (year >= -509 && year < -264) {
    L.polygon(romanItaly, { color: '#9333ea', weight: 2, fillOpacity:
0.35 }).addTo(empireLayer).bindTooltip("<b>羅馬共和國 (早期)</b>", {sticky: true});
    } else if (year >= -264 && year < -146) {
        L.polygon(romanPunic, { color: '#9333ea', weight: 2, fillOpacity:
0.35 }).addTo(empireLayer).bindTooltip("<b>羅馬共和國 (布匿戰爭)</b>", {sticky:
true});
    } else if (year >= -146 && year < -27) {
        L.polygon(romanLateRep, { color: '#9333ea', weight: 2, fillOpacity:
0.35 }).addTo(empireLayer).bindTooltip("<b>羅馬共和國 (晚期)</b>", {sticky: true});
    } else if (year >= -27 && year < 117) {
        L.polygon(romanEarlyEmp, { color: '#dc2626', weight: 2, fillOpacity:
0.35 }).addTo(empireLayer).bindTooltip("<b>羅馬帝國 (早期)</b>", {sticky: true});
    } else if (year >= 117 && year < 395) {

```

```

        L.polygon(romanPeakEmp, { color: '#dc2626', weight: 2, fillOpacity:
0.35 }).addTo(empireLayer).bindTooltip("<b>大羅馬帝國 (極盛期)</b>", {sticky: true});
    } else if (year >= 395 && year < 476) {
        L.polygon(westernRomanCoords, { color: '#2563eb', weight: 2,
fillOpacity: 0.35 }).addTo(empireLayer).bindTooltip("<b>西羅馬帝國</b>", {sticky: true});
        L.polygon(easternRomanCoords, { color: '#d97706', weight: 2,
fillOpacity: 0.4 }).addTo(empireLayer).bindTooltip("<b>東羅馬帝國</b>", {sticky: true});
    } else if (year >= 476 && year < 534) {
        L.polygon(easternRomanCoords, { color: '#d97706', weight: 2,
fillOpacity: 0.4 }).addTo(empireLayer).bindTooltip("<b>東羅馬帝國</b>", {sticky: true});
    } else if (year >= 534 && year < 650) {
        L.polygon(justinianEmpireCoords, { color: '#d97706', weight: 2,
fillOpacity: 0.4 }).addTo(empireLayer).bindTooltip("<b>東羅馬帝國 (查士丁尼時
期)</b>", {sticky: true});
    } else if (year >= 650 && year < 1071) {
        L.polygon(byzantineShrunkCoords, { color: '#d97706', weight: 2,
fillOpacity: 0.4 }).addTo(empireLayer).bindTooltip("<b>拜占庭帝國 (中世紀)</b>",
{sticky: true});
    } else if (year >= 1071 && year <= 1453) {
        L.polygon(byzantineLateCoords, { color: '#d97706', weight: 2,
fillOpacity: 0.4 }).addTo(empireLayer).bindTooltip("<b>拜占庭帝國 (晚期)</b>", {sticky:
true});
    }

    if (year >= 661 && year <= 1258) {
        L.polygon(arabEmpireCoords, { color: '#16a34a', weight: 2, fillOpacity:
0.3 }).addTo(empireLayer).bindTooltip("<b>阿拉伯帝國</b>", {sticky: true});
    }

    if (year >= 1206 && year < 1271) {
        L.polygon(mongolEmpireCoords, { color: '#1d4ed8', weight: 3,
fillOpacity: 0.25, dashArray: '10, 10' }).addTo(empireLayer).bindTooltip("<b>大蒙古國
</b>", {sticky: true});
    } else if (year >= 1271 && year < 1368) {

```

```

        L.polygon(yuanDynastyCoords, { color: '#eab308', weight: 2, fillOpacity:
0.35 }).addTo(empireLayer).bindTooltip("<b>大元 (元朝)</b>", {sticky: true});

        L.polygon(mongolEmpireCoords, { color: '#1d4ed8', weight: 3,
fillOpacity: 0.1, dashArray: '10, 10' }).addTo(empireLayer).bindTooltip("<b>四大汗國與泛
蒙古勢力範圍</b>", {sticky: true});

        } else if (year >= 1368 && year < 1644) {
            if (year >= 1402 && year <= 1435) {
                L.polygon(mingYongleCoords, { color: '#d97706', weight: 2,
fillOpacity: 0.35 }).addTo(empireLayer).bindTooltip("<b>大明 (永樂盛世：含交趾與奴兒
千都司)</b>", {sticky: true});
            } else {
                L.polygon(mingRegularCoords, { color: '#d97706', weight: 2,
fillOpacity: 0.35 }).addTo(empireLayer).bindTooltip("<b>大明 (明朝)</b>", {sticky: true});
            }
        } else if (year >= 1644 && year < 1912) {
            if (year < 1759) {
                L.polygon(qingEarlyCoords, { color: '#f59e0b', weight: 2,
fillOpacity: 0.3 }).addTo(empireLayer).bindTooltip("<b>大清帝國 (擴張期)</b>", {sticky:
true});
            } else if (year >= 1759 && year < 1860) {
                L.polygon(qingPeakCoords, { color: '#f59e0b', weight: 2,
fillOpacity: 0.35 }).addTo(empireLayer).bindTooltip("<b>大清帝國 (極盛期：秋海棠版
圖)</b>", {sticky: true});
            } else {
                L.polygon(qingLateCoords, { color: '#f59e0b', weight: 2, fillOpacity:
0.35 }).addTo(empireLayer).bindTooltip("<b>晚清疆域 (1860 年後割讓外滿洲)</b>",
{sticky: true});
            }
        } else if (year >= 1912 && year < 1949) {
            L.polygon(rocCoords, { color: '#3b82f6', weight: 2, fillOpacity:
0.25 }).addTo(empireLayer).bindTooltip("<b>中華民國 (包含蒙古與西藏等名義宣
稱)</b>", {sticky: true});
            if (year >= 1932 && year < 1945) {

```

```

        L.polygon(manchukuoCoords, { color: '#ef4444', weight: 2,
fillOpacity: 0.4, dashArray: '5,5' }).addTo(empireLayer).bindTooltip("<b>滿洲國（日本扶植
之傀儡政權）</b>", {sticky: true});
    }
    if (year >= 1937 && year < 1945) {
        L.polygon(japaneseOccupiedChina, { color: '#ef4444', weight: 2,
fillOpacity: 0.3, dashArray: '5,5' }).addTo(empireLayer).bindTooltip("<b>日本淪陷區（全面
抗戰時期）</b>", {sticky: true});
    }
    } else if (year >= 1949) {
        L.polygon(prcCoords, { color: '#ef4444', weight: 2, fillOpacity:
0.35 }).addTo(empireLayer).bindTooltip("<b>中華人民共和國</b>", {sticky: true});
    }

    if (year >= -1550 && year <= -1070) {
        L.polygon(egyptNewKingdomCoords, { color: '#eab308', weight: 2,
fillOpacity: 0.35 }).addTo(empireLayer).bindTooltip("<b>古埃及新王國時期</b>", {sticky:
true});
    }
    if (year >= -334 && year <= -300) {
        L.polygon(alexanderEmpireCoords, { color: '#0ea5e9', weight: 2,
fillOpacity: 0.35 }).addTo(empireLayer).bindTooltip("<b>亞歷山大帝國</b>", {sticky:
true});
    }

    // 中國歷代疆域變遷
    if (year >= -2070 && year < -1600) {
        L.polygon(xiaDynastyCoords, { color: '#f59e0b', weight: 2, fillOpacity:
0.3 }).addTo(empireLayer).bindTooltip("<b>夏朝</b>", {sticky: true});
    } else if (year >= -1600 && year < -1046) {
        L.polygon(shangDynastyCoords, { color: '#d97706', weight: 2,
fillOpacity: 0.3 }).addTo(empireLayer).bindTooltip("<b>商朝</b>", {sticky: true});
    } else if (year >= -1046 && year < -770) {

```

```

        L.polygon(westernZhouCoords, { color: '#b45309', weight: 2, fillOpacity:
0.3 }).addTo(empireLayer).bindTooltip("<b>西周</b>", {sticky: true});
    } else if (year >= -770 && year < -221) {
        L.polygon(easternZhouCoords, { color: '#92400e', weight: 2, fillOpacity:
0.3, dashArray: '5,5' }).addTo(empireLayer).bindTooltip("<b>東周 (春秋戰國)</b>",
{sticky: true});
    } else if (year >= -221 && year < -202) {
        L.polygon(qinDynastyCoords_ancient, { color: '#000000', weight: 2,
fillOpacity: 0.4 }).addTo(empireLayer).bindTooltip("<b>秦朝</b>", {sticky: true});
    } else if (year >= -202 && year < -141) {
        L.polygon(hanEarlyCoords, { color: '#ef4444', weight: 2, fillOpacity:
0.3 }).addTo(empireLayer).bindTooltip("<b>西漢 (初期)</b>", {sticky: true});
    } else if (year >= -141 && year < 8) {
        L.polygon(hanWuCoords, { color: '#dc2626', weight: 2, fillOpacity:
0.35 }).addTo(empireLayer).bindTooltip("<b>西漢 (武帝擴張後/全盛)</b>", {sticky:
true});
    } else if (year >= 25 && year < 220) {
        L.polygon(easternHanCoords, { color: '#dc2626', weight: 2, fillOpacity:
0.35 }).addTo(empireLayer).bindTooltip("<b>東漢</b>", {sticky: true});
    } else if (year >= 220 && year < 280) {
        L.polygon(threeKingdomsWei, { color: '#eab308', weight: 2, fillOpacity:
0.35 }).addTo(empireLayer).bindTooltip("<b>曹魏</b>", {sticky: true});
        L.polygon(threeKingdomsShu, { color: '#ef4444', weight: 2, fillOpacity:
0.35 }).addTo(empireLayer).bindTooltip("<b>蜀漢</b>", {sticky: true});
        L.polygon(threeKingdomsWu, { color: '#3b82f6', weight: 2, fillOpacity:
0.35 }).addTo(empireLayer).bindTooltip("<b>孫吳</b>", {sticky: true});
    } else if (year >= 280 && year < 316) {
        L.polygon(westernJinCoords, { color: '#9333ea', weight: 2, fillOpacity:
0.35 }).addTo(empireLayer).bindTooltip("<b>西晉</b>", {sticky: true});
    } else if (year >= 317 && year < 420) {
        L.polygon(easternJinCoords, { color: '#ef4444', weight: 2, fillOpacity:
0.35 }).addTo(empireLayer).bindTooltip("<b>東晉 (衣冠南渡)</b>", {sticky: true});

```

```

        L.polygon(sixteenKingdomsCoords, { color: '#3b82f6', weight: 2,
fillOpacity: 0.35, dashArray: '5,5' }).addTo(empireLayer).bindTooltip("<b>五胡十六國</b>", {sticky: true});
    } else if (year >= 420 && year < 589) {
        L.polygon(southernDynasties, { color: '#ef4444', weight: 2, fillOpacity:
0.35 }).addTo(empireLayer).bindTooltip("<b>南朝 (宋、齊、梁、陳)</b>", {sticky:
true});
        L.polygon(northernDynasties, { color: '#3b82f6', weight: 2, fillOpacity:
0.35 }).addTo(empireLayer).bindTooltip("<b>北朝 (北魏、東/西魏、北周/北齊)</b>",
{sticky: true});
    } else if (year >= 589 && year < 618) {
        L.polygon(suiDynastyCoords, { color: '#f59e0b', weight: 2, fillOpacity:
0.35 }).addTo(empireLayer).bindTooltip("<b>隋朝</b>", {sticky: true});
    } else if (year >= 618 && year < 907) {
        if (year < 630) {
            L.polygon(tangEarlyCoords, { color: '#f97316', weight: 2,
fillOpacity: 0.3 }).addTo(empireLayer).bindTooltip("<b>唐朝 (初唐)</b>", {sticky: true});
        } else if (year >= 630 && year < 755) {
            L.polygon(tangPeakCoords, { color: '#ea580c', weight: 2,
fillOpacity: 0.35 }).addTo(empireLayer).bindTooltip("<b>唐朝 (極盛期 / 天可汗體
系)</b>", {sticky: true});
        }
        L.polygon(tuboEmpireCoords, { color: '#9333ea', weight: 2,
fillOpacity: 0.3 }).addTo(empireLayer).bindTooltip("<b>吐蕃帝國</b>", {sticky: true});
    } else {
        L.polygon(tangLateCoords, { color: '#f97316', weight: 2, fillOpacity:
0.25 }).addTo(empireLayer).bindTooltip("<b>唐朝 (中晚唐 / 藩鎮割據)</b>", {sticky:
true});
        L.polygon(tuboMaxCoords, { color: '#9333ea', weight: 2,
fillOpacity: 0.35 }).addTo(empireLayer).bindTooltip("<b>吐蕃帝國 (擴張至河西走
廊)</b>", {sticky: true});
        L.polygon(uyghurCoords, { color: '#0ea5e9', weight: 2, fillOpacity:
0.3 }).addTo(empireLayer).bindTooltip("<b>回鶻汗國</b>", {sticky: true});
    }
}

```



```

    } else if (year >= 907 && year < 960) {
        L.polygon(fiveDynastiesCoords, { color: '#9ca3af', weight: 2, fillOpacity:
0.3 }).addTo(empireLayer).bindTooltip("<b>五代十國</b>", {sticky: true});
        if (year >= 916) {
            L.polygon(liaoDynastyCoords, { color: '#6b7280', weight: 2,
fillOpacity: 0.3 }).addTo(empireLayer).bindTooltip("<b>遼國 (契丹)</b>", {sticky: true});
        }
    } else if (year >= 960 && year < 1127) {
        L.polygon(northernSongCoords, { color: '#dc2626', weight: 2,
fillOpacity: 0.3 }).addTo(empireLayer).bindTooltip("<b>北宋</b>", {sticky: true});
        L.polygon(liaoDynastyCoords, { color: '#6b7280', weight: 2, fillOpacity:
0.3 }).addTo(empireLayer).bindTooltip("<b>遼國 (契丹)</b>", {sticky: true});
        if (year >= 1038) {
            L.polygon(westernXiaCoords, { color: '#8b5cf6', weight: 2,
fillOpacity: 0.35 }).addTo(empireLayer).bindTooltip("<b>西夏 (党項)</b>", {sticky: true});
        }
    } else if (year >= 1127 && year < 1279) {
        L.polygon(southernSongCoords, { color: '#dc2626', weight: 2,
fillOpacity: 0.3 }).addTo(empireLayer).bindTooltip("<b>南宋</b>", {sticky: true});
        if (year < 1234) {
            L.polygon(jurchenJinCoords, { color: '#eab308', weight: 2,
fillOpacity: 0.3 }).addTo(empireLayer).bindTooltip("<b>金國 (女真)</b>", {sticky: true});
        }
        if (year < 1227) {
            L.polygon(westernXiaCoords, { color: '#8b5cf6', weight: 2,
fillOpacity: 0.35 }).addTo(empireLayer).bindTooltip("<b>西夏 (党項)</b>", {sticky: true});
        }
    }

    if (year >= 1521 && year <= 1825) {
        L.polygon(spanishEmpireCoords, { color: '#eab308', weight: 2,
fillOpacity: 0.3 }).addTo(empireLayer).bindTooltip("<b>西班牙帝國 (海上霸權)</b>",
{sticky: true});
    }

```

```

    }
    if (year >= 1858 && year <= 1947) {
        L.polygon(britishEmpireCoords, { color: '#ec4899', weight: 2,
fillOpacity: 0.3 }).addTo(empireLayer).bindTooltip("<b>大英帝國（日不落帝國）</b>",
{sticky: true});
    }
    if (year >= 1939 && year <= 1945) {
        L.polygon(axisCoords, { color: '#000000', weight: 2, fillOpacity: 0.5,
dashArray: '5,5' }).addTo(empireLayer).bindTooltip("<b>軸心國勢力範圍（二戰）</b>",
{sticky: true});
    }
    if (year >= 1922 && year <= 1991) {
        L.polygon(ussrCoords, { color: '#dc2626', weight: 2, fillOpacity:
0.25 }).addTo(empireLayer).bindTooltip("<b>蘇維埃社會主義共和國聯盟（蘇聯）</b>",
{sticky: true});
    }

    // 事件星標（±150 年內）—— 選中的事件用不同外觀突顯
    // ✨ 安全過濾：確保所有座標有效再生成，避免 L.marker 崩潰
    const activeEvents = historyData.filter(ev =>
        ev && Number.isFinite(ev.lat) && Number.isFinite(ev.lng) &&
        Math.abs(ev.year - year) <= 150 && activeTracks.includes(ev.track)
    );

    activeEvents.forEach(event => {
        const displayYearText = event.year < 0 ? `前 ${Math.abs(event.year)}` :
`西元 ${event.year}`;
        const isSel = event.id === selectedEventId;
        const icon = L.divIcon({
            className: 'custom-div-icon',
            html: isSel

```

```

? `<div class="w-8 h-8 rounded-full bg-white border-2 border-
primary flex items-center justify-center shadow-[0_0_22px_rgba(212,175,55,1)] cursor-
pointer"><i class="fa-solid fa-star text-primary text-sm"></i></div>`
: `<div class="w-6 h-6 rounded-full bg-primary border-2
border-white flex items-center justify-center shadow-[0_0_15px_rgba(212,175,55,1)]
animate-bounce cursor-pointer"><i class="fa-solid fa-star text-dark text-[10px]"></i></div>`,
iconSize: isSel ? [32, 32] : [24, 24], iconAnchor: isSel ? [16, 16] :
[12, 12]

});

const marker = L.marker([event.lat, event.lng], { icon: icon,
zIndexOffset: isSel ? 2000 : 1000 }).addTo(map);

marker.bindTooltip(`<div class="bg-gray-900 border border-primary p-
1.5 rounded text-xs font-bold text-white">${displayYearText} 年 : ${event.title}</div>`,
{permanent: false, direction: 'top'});

marker.on('click', () => selectEvent(event.id));

markers.push(marker);

});

}

//

=====

=

// AI 探究引擎

//

=====

=

const personas = {

'socratic': { name: '蘇格拉底歷史教練', systemPrompt: `你是一個運用「蘇
格拉底反詰法」的歷史導師。引導高中生思考歷史因果，而不是直接給答案。當使用
者提出觀點時，請回問 1-2 個能刺激他們反思歷史脈絡、資源分配、統治權威合法性
的開放式問題。語氣要有智慧、親切，使用現代繁體中文。`},

'roleplay': { name: '歷史關鍵人物 (動態扮演)', systemPrompt: `你現在正在
進行歷史角色扮演(Cosplay)。你需要根據使用者傳入的歷史事件，自動切換成當時最

```

關鍵的統治者、領袖或歷史人物。請先表明你的歷史身分，隨後完全融入該人物性格，並提出需要使用者抉擇的歷史難題。使用繁體中文。`},

```
'debate': { name: '歷史爭議辯論庭主持人', systemPrompt: `你是一個客觀嚴謹的歷史辯論法官。針對具爭議的歷史事件，你需要主持辯論。請扮演使用者的「反方」提出有力的反駁，要求其從「歷史正義」、「階級局限」或「當代普世價值」等不同角度進行交叉質詢。使用繁體中文。` }
```

```
};
```

```
function setAIPersona(personaKey, isInit = false) {
    currentPersona = personas[personaKey];
    document.getElementById('chat-title').innerHTML = `當前模式：<span
class="text-primary">${currentPersona.name}</span>`;
    document.querySelectorAll('.persona-btn').forEach(btn =>
btn.classList.remove('border-primary', 'bg-gray-700'));
    const activeBtn = document.querySelector(`.persona-
btn[onclick="setAIPersona('${personaKey}')]`);
    if (activeBtn) activeBtn.classList.add('border-primary', 'bg-gray-700');

    if (isInit) {
        aiChatHistory = [];
        document.getElementById('chat-box').innerHTML = "";
        appendMessage('AI', `系統已成功載入【${currentPersona.name}】設
定。請從時空地圖點擊事件，選擇想要的探究模式，或直接輸入你想探討的議題。`);
        if (!apiKey) document.getElementById('ai-key-
note').classList.remove('hidden');
    } else {
        appendSystemMessage(`  已切換至【${currentPersona.name}】模
式。將延續剛才的歷史對話脈絡。`);
    }
}
```

```
function appendSystemMessage(text) {
    const chatBox = document.getElementById('chat-box');
```

```

const msgDiv = document.createElement('div');
msgDiv.className = 'flex justify-center my-3 fade-in';
msgDiv.innerHTML = `<div class="text-xs text-gray-400 bg-gray-800/80 px-4
py-1.5 rounded-full shadow-inner border border-gray-700">${text}</div>`;
chatBox.appendChild(msgDiv);
chatBox.scrollTop = chatBox.scrollHeight;
aiChatHistory.push(`(System: 使用者已將探究模式切換為
${currentPersona.name})`);
}

```

```

function exploreInAI(mode) {
  if(!selectedEvent) return;
  const topic = selectedEvent.title;
  const yearStr = selectedEvent.year < 0 ? `西元前
${Math.abs(selectedEvent.year)}` : `西元 ${selectedEvent.year}`;
  closeEventPanel();
  switchTab('ai-view');
  setAIPersona(mode);
  let initialPrompt = "";
  if (mode === 'socratic') initialPrompt = `我想探討 ${yearStr} 年的
「${topic}」。請用蘇格拉底提問法引導我思考這個事件的歷史脈絡與深遠影響。`;
  else if (mode === 'roleplay') initialPrompt = `我想回到 ${yearStr} 年探討
「${topic}」。請完全扮演當時的關鍵歷史人物與我展開對話，並拋出你需要我做出的
決策難題。`;
  else if (mode === 'debate') initialPrompt = `我想針對 ${yearStr} 年的
「${topic}」進行歷史辯證。請扮演反方或法官，針對此事件的爭議點向我提出嚴厲的
質詢。`;
  document.getElementById('ai-input').value = initialPrompt;
  sendAIMessage();
}

```

```

function appendMessage(sender, text) {
  const chatBox = document.getElementById('chat-box');

```

```

const msgDiv = document.createElement('div');
if (sender === 'User') {
    msgDiv.className = 'flex justify-end fade-in';
    msgDiv.innerHTML = `<div class="max-w-[80%] chat-bubble-user p-4
rounded-xl rounded-tr-none text-gray-100 shadow-md text-[14px] leading-
relaxed">${escapeHtml(text).replace(/\n/g, '<br>')}</div>`;
} else {
    msgDiv.className = 'flex justify-start fade-in';
    msgDiv.innerHTML = `<div class="max-w-[80%] chat-bubble-ai p-4
rounded-xl rounded-tl-none text-gray-100 shadow-md flex gap-3 text-[14px] leading-
relaxed"><i class="fa-solid fa-robot mt-1 text-primary shrink-0 text-
lg"></i><div>${escapeHtml(text).replace(/\n/g, '<br>')}</div></div>`;
}
chatBox.appendChild(msgDiv);
chatBox.scrollTop = chatBox.scrollHeight;
}

async function fetchGeminiWithRetry(prompt, systemInstruction, retries = 5) {
    const delays = [1000, 2000, 4000, 8000, 16000];
    const payload = { contents: [{ parts: [{ text: prompt }] }], systemInstruction:
    { parts: [{ text: systemInstruction }] } };
    for (let i = 0; i < retries; i++) {
        try {
            const response = await
            fetch(`https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash-preview-
09-2025:generateContent?key=${apiKey}`, {
                method: 'POST', headers: { 'Content-Type': 'application/json' },
                body: JSON.stringify(payload)
            });
            if (!response.ok) throw new Error(`HTTP error! status:
${response.status}`);
            const data = await response.json();

```

```
        return data.candidates?.[0]?.content?.parts?.[0]?.text || "抱歉，我無法產出回應。";
```

```
    } catch (error) {  
      if (i === retries - 1) throw error;  
      await new Promise(res => setTimeout(res, delays[i]));  
    }  
  }  
}
```

```
async function sendAIMessage() {  
  if (!currentPersona) return;  
  const inputEl = document.getElementById('ai-input');  
  const text = inputEl.value.trim();  
  if (!text) return;
```

```
  appendMessage('User', text);  
  inputEl.value = "";  
  aiChatHistory.push(`User: ${text}`);
```

```
  if (!apiKey) {  
    appendMessage('AI', '⚠️ 尚未設定 API 金鑰。此頁面在 Gemini Canvas / AI Studio 環境會自動注入；若獨立開啟，請在程式碼頂端的 apiKey 填入你的 Gemini API Key。');  
    return;
```

```
  }  
  
  const conversationContext = aiChatHistory.slice(-20).join("\n");  
  document.getElementById('ai-loading').classList.remove('hidden');
```

```
  try {  
    const prompt = `目前的對話紀錄：\n${conversationContext}\n\n請根據你的角色設定，接續上述對話給予自然、深刻且口語化的回應。切勿使用死板的模板，請針對 User 的最新提問或回答進行針對性的互動與延伸。`;
```

```

        const aiResponse = await fetchGeminiWithRetry(prompt,
currentPersona.systemPrompt);

        document.getElementById('ai-loading').classList.add('hidden');
        appendMessage('AI', aiResponse);
        aiChatHistory.push(`AI: ${aiResponse}`);
    } catch (error) {
        document.getElementById('ai-loading').classList.add('hidden');
        appendMessage('AI', '⚠️ 連線發生錯誤，請稍後再試。');
    }
}

//
=====
=

// 歷史沙龍 —— 本地身分系統 + localStorage 持久化 + 連結地圖事件
//
=====
=

const AVATAR_COLORS = ['bg-blue-600','bg-red-600','bg-green-600','bg-purple-
600','bg-pink-600','bg-amber-600','bg-teal-600','bg-indigo-600'];

let userIdentity = null;
let pendingIdentityAction = null;
let forumPosts = [];
let currentOpenPostId = null;

function colorForName(name) {
    let h = 0;
    for (let i = 0; i < name.length; i++) h = (h * 31 + name.charCodeAt(i)) >>> 0;
    return AVATAR_COLORS[h % AVATAR_COLORS.length];
}

function loadIdentity() {
    const raw = store.get('hlab_identity');
    if (raw) { try { return JSON.parse(raw); } catch(e) {} }

```



```

        return null;
    }
    function saveIdentity(id) {
        userIdentity = id;
        store.set('hlab_identity', JSON.stringify(id));
        renderIdentityBar();
    }
    function renderIdentityBar() {
        const bar = document.getElementById('identity-bar');
        if (!bar) return;
        if (userIdentity && userIdentity.name) {
            bar.innerHTML = `
                <div class="w-7 h-7 rounded-full ${userIdentity.color} flex items-
center justify-center text-white text-xs font-bold shadow-
inner">${escapeHtml(userIdentity.name.charAt(0).toUpperCase())}</div>
                <span class="text-sm text-gray-200 font-medium max-w-[120px]
truncate">${escapeHtml(userIdentity.name)}</span>
                <button onclick="openIdentityModal()" class="text-xs text-gray-400
hover:text-primary underline ml-1">更換</button>`;
        } else {
            bar.innerHTML = `
                <i class="fa-regular fa-circle-user text-gray-500"></i>
                <span class="text-sm text-gray-500">訪客（未設定身分）</span>
                <button onclick="openIdentityModal()" class="text-xs text-primary
hover:underline ml-1 font-bold">設定身分</button>`;
        }
    }
    function openIdentityModal(onDone = null) {
        pendingIdentityAction = (typeof onDone === 'function') ? onDone : null;
        document.getElementById('identity-name-input').value = userIdentity?.name ||
";

        const m = document.getElementById('identity-modal');
        m.classList.remove('hidden'); m.classList.add('flex');
    }

```

```

        setTimeout(() => document.getElementById('identity-name-input').focus(),
50);
    }
    function closeIdentityModal() {
        pendingIdentityAction = null;
        closeModal('identity-modal');
    }
    function saveIdentityFromModal() {
        const name = document.getElementById('identity-name-input').value.trim();
        if (!name) { showToast('請先輸入一個名字'); return; }
        saveIdentity({ name: name, color: colorForName(name) });
        closeModal('identity-modal');
        showToast('✅ 已設定身分：' + name);
        const cb = pendingIdentityAction; pendingIdentityAction = null;
        if (cb) cb();
    }
    function requireIdentity(next) {
        if (userIdentity && userIdentity.name) { next(); }
        else { showToast('發言前，請先設定你在沙龍中的身分');
openIdentityModal(next); }
    }

    function seedPosts() {
        return [{
            id: 'seed_hammurabi',
            title: '如何評價《漢穆拉比法典》中的「以眼還眼」原則？',
            content: '法典中明文規定了同態復仇，但如果仔細閱讀，會發現處罰
嚴重程度完全取決於階級（被害人是奴隸或貴族）。這到底是法治的萌芽，還是統治
階級鞏固特權的工具？',
            author: 'Hammurabi_Critic', avatarColor: 'bg-blue-600',
            tag: '古代法制', tagBg: 'bg-red-900', tagText: 'text-red-200',
            timeText: '3 小時前',

```

```
      comments: [{ author: '歷史思辨學者', color: 'text-primary', text: '這是一個非常好的切入點！讓我們能穿透文獻表面，探究統治階層底層的權力架構，這正是歷史思辨的價值。', timeText: '2 小時前' }]
```

```
    }];
```

```
  }
```

```
  function loadForumPosts() {
```

```
    const raw = store.get('hlab_forum');
```

```
    if (raw) { try { const a = JSON.parse(raw); if (Array.isArray(a) && a.length
```

```
return a; } catch(e){} }
```

```
    return seedPosts();
```

```
  }
```

```
  function saveForumPosts() { store.set('hlab_forum', JSON.stringify(forumPosts)); }
```

```
  function renderForum() {
```

```
    const grid = document.getElementById('forum-grid');
```

```
    grid.innerHTML = "";
```

```
    forumPosts.forEach(p => {
```

```
      const card = document.createElement('div');
```

```
      card.className = 'bg-panel p-5 rounded-xl border border-gray-800
```

```
hover:border-gray-500 transition cursor-pointer shadow-lg fade-in';
```

```
      card.onclick = () => openForumPost(p.id);
```

```
      const linked = p.eventId ? `<span class="inline-block text-[10px] text-  
primary border border-primary/50 rounded px-1.5 py-0.5 ml-1 align-middle"><i class="fa-  
solid fa-location-dot mr-1"></i>連結地圖事件</span>` : "";
```

```
      card.innerHTML = `
```

```
        <div class="flex justify-between items-start mb-3">
```

```
          <span class="${p.tagBg} ${p.tagText} text-xs px-2 py-1  
rounded">${escapeHtml(p.tag)}</span>
```

```
          <span class="text-gray-500 text-  
xs">${escapeHtml(p.timeText)}</span>
```

```
        </div>
```

```
        <h3 class="text-lg font-bold text-gray-200 mb-  
2">${escapeHtml(p.title)}${linked}</h3>
```

```

        <p class="text-gray-400 text-sm mb-4 line-clamp-
3">${escapeHtml(p.content)}</p>

        <div class="flex items-center justify-between text-gray-500 text-
sm">

            <div class="flex items-center gap-2">

                <div class="w-6 h-6 rounded-full ${p.avatarColor} flex
items-center justify-center text-white text-xs shadow-
inner">${escapeHtml(p.author.charAt(0).toUpperCase())}</div>

                <span>${escapeHtml(p.author)}</span>

            </div>

            <div><i class="fa-regular fa-comment-dots mr-1 text-
primary"></i> ${p.comments.length} 則思辨</div>

        </div>`;

        grid.appendChild(card);

    });

}

function renderComments(p) {

    const c = document.getElementById('mock-comments');

    if (!p.comments.length) {

        c.innerHTML = `<div class="text-sm text-gray-500 italic text-center py-
4">尚無留言，成為第一個發表思辨的人。</div>`;

        return;

    }

    c.innerHTML = p.comments.map(cm => `

        <div class="bg-panel p-4 rounded-lg border border-gray-800 shadow-sm
${cm.mine ? 'border-l-4 border-l-green-500' : ''}>

            <div class="flex justify-between items-center mb-2">

                <div class="text-xs font-bold ${cm.color} || 'text-
primary'">${escapeHtml(cm.author)}${cm.mine ? ' (您)' : ''}</div>

                <div class="text-xs text-gray-
500">${escapeHtml(cm.timeText)}</div>

            </div>

```

```
        <p class="text-sm text-gray-200 leading-relaxed">${escapeHtml(cm.text).replace(/\n/g, '<br>')}</p>
```

```
    </div>`).join("");
```

```
}
```

```
function openForumPost(postId) {
```

```
    const p = forumPosts.find(x => x.id === postId);
```

```
    if (!p) return;
```

```
    currentOpenPostId = postId;
```

```
    document.getElementById('read-post-title').innerText = p.title;
```

```
    document.getElementById('read-post-content').innerText = p.content;
```

```
    document.getElementById('read-post-author').innerText = p.author;
```

```
    const av = document.getElementById('read-post-avatar');
```

```
    av.innerText = p.author.charAt(0).toUpperCase();
```

```
    av.className = `w-6 h-6 rounded-full ${p.avatarColor} flex items-center justify-center text-white text-xs shadow-inner`;
```

```
    document.getElementById('read-post-time').innerText = p.timeText;
```

```
    const tagEl = document.getElementById('read-post-tag');
```

```
    tagEl.innerText = p.tag;
```

```
    tagEl.className = `text-xs px-2 py-1 rounded mb-3 inline-block ${p.tagBg} ${p.tagText}`;
```

```
    document.getElementById('forum-reply-input').value = "";
```

```
    renderComments(p);
```

```
    const m = document.getElementById('forum-read-modal');
```

```
    m.classList.remove('hidden'); m.classList.add('flex');
```

```
}
```

```
function openNewPostModal() {
```

```
    requireIdentity(() => {
```

```
        document.getElementById('new-post-title').value = "";
```

```
        document.getElementById('new-post-content').value = "";
```

```
        const m = document.getElementById('forum-write-modal');
```

```
        m.classList.remove('hidden'); m.classList.add('flex');
```

```
    });  
  }  
}
```

```
function submitNewPost() {  
  const title = document.getElementById('new-post-title').value.trim();  
  const content = document.getElementById('new-post-content').value.trim();  
  const tag = document.getElementById('new-post-tag').value;  
  if (!title || !content) { showToast("標題與內容不能留白！"); return; }  
  requireIdentity(() => {  
    const tagColors = [{bg:'bg-blue-900',text:'text-blue-200'}, {bg:'bg-yellow-900',text:'text-yellow-200'}, {bg:'bg-green-900',text:'text-green-200'}, {bg:'bg-purple-900',text:'text-purple-200'}];  
    const tc = tagColors[Math.floor(Math.random() * tagColors.length)];  
    const post = {  
      id: 'post_' + Date.now(),  
      title, content, tag,  
      author: userIdentity.name, avatarColor: userIdentity.color,  
      tagBg: tc.bg, tagText: tc.text, timeText: '剛剛', comments: []  
    };  
    forumPosts.unshift(post);  
    saveForumPosts();  
    renderForum();  
    closeModal('forum-write-modal');  
    showToast("✅ 新議題發布成功！已加入歷史沙龍。");  
  });  
}
```

```
function submitForumReply() {  
  const inputEl = document.getElementById('forum-reply-input');  
  const text = inputEl.value.trim();  
  if (!text) { showToast("回覆內容不能為空！"); return; }  
  requireIdentity(() => {  
    const p = forumPosts.find(x => x.id === currentOpenPostId);
```

```

        if (!p) return;
        p.comments.push({ author: userIdentity.name, color: 'text-green-400',
text: text, timeText: '剛剛', mine: true });
        saveForumPosts();
        renderComments(p);
        renderForum(); // 更新卡片上的留言數
        inputEl.value = "";
        const modalBox = document.querySelector('#forum-read-modal > div');
        modalBox.scrollTo({ top: modalBox.scrollHeight, behavior: 'smooth' });
        showToast("✅ 回覆已發布！");
    });
}

function discussSelectedEvent() {
    if (!selectedEvent) return;
    const event = selectedEvent;
    requireIdentity(() => {
        let post = forumPosts.find(p => p.eventId === event.id);
        if (!post) {
            const yearStr = event.year < 0 ? `西元前 ${Math.abs(event.year)}` :
`西元 ${event.year}`;
            post = {
                id: 'evt_' + event.id,
                title: `【地圖事件討論】${event.title}`,
                content: `針對 ${yearStr} 年的「${event.title}」展開討論。
\n\n 事件背景：${event.desc}`,
                author: '時空地圖', avatarColor: 'bg-primary',
                tag: event.track, tagBg: 'bg-gray-700', tagText: 'text-gray-200',
                timeText: '由地圖建立', comments: [], eventId: event.id
            };
            forumPosts.unshift(post);
            saveForumPosts();
            renderForum();
        }
    });
}

```

```

        }
        closeEventPanel();
        switchTab('social-view');
        openForumPost(post.id);
    });
}

//
=====

=

    // 事件綁定與啟動
    //
=====

=

    document.getElementById('forum-reply-input').addEventListener('keydown',
function(e) {
    if (e.key === 'Enter' && !e.shiftKey) { e.preventDefault();
submitForumReply(); }
    });
    document.getElementById('identity-name-input').addEventListener('keydown',
function(e) {
    if (e.key === 'Enter') { e.preventDefault(); saveIdentityFromModal(); }
    });
    document.getElementById('ai-input').addEventListener('keydown', function(e) {
    if (e.key === 'Enter' && !e.shiftKey) { e.preventDefault();
sendAIMessage(); }
    });

    window.onload = () => {
        userIdentity = loadIdentity();
        forumPosts = loadForumPosts();
        renderIdentityBar();
        renderForum();
    }

```



```
    updateTrackFilterUI();  
    initMap();  
    setAIPersona('socratic', true);  
    setTimeout(() => { map.invalidateSize(); }, 500);  
  };  
  window.addEventListener('resize', () => { if (map) map.invalidateSize(); });  
</script>  
</body>  
</html>
```